Amplitude Selection, V1 Connectivity, and Numerical Tests for the Spatio-temporal Hallucination Model

August 17, 2026

1 Main conclusions

The two critical waves have amplitudes A and B. Without forcing their cubic equations are

$$\dot{A} = A(a + b|A|^2 + c|B|^2), \qquad \dot{B} = B(a + b|B|^2 + c|A|^2). \tag{1}$$

Here and below the dot denotes the slow time. There are only three generic amplitude states: rest, a traveling wave, and a standing wave. Their existence and stability follow from two 2×2 Jacobians and are derived in Section 2.

Suppose each separated forcing is either absent or has order ϵ^2, with temporal frequency $2\omega_0$ and spatial wavenumber $2k_0$. A direct projection and a parity argument show that no additive critical forcing is produced: the reduced coefficient previously denoted by f is identically zero in this regime. After choosing the temporal and spatial origins, the two remaining parametric strengths are real and nonnegative:

$$\begin{aligned} \dot{A} &= A(a + b|A|^2 + c|B|^2) + \eta_t \overline{B} + \eta_x B, \\ \dot{B} &= B(a + b|B|^2 + c|A|^2) + \eta_t \overline{A} + \eta_x A. \end{aligned} \tag{2}$$

Section 3 derives (2) from the forcing orders, reconstructs every locked phase from the two amplitudes, reduces the equal-amplitude branches to a quadratic in the squared amplitude, reduces every unequal-amplitude branch to one scalar equation, and gives exact stability tests for the rest state, every equal branch, and every remaining complex equilibrium.

For K cell types, let $\boldsymbol{\mu} = (\mu_{ij})$ be the signed Gaussian masses and $\Sigma = (\sigma_{ij})$ the widths. The exact Fourier connectivity is $\widehat{J}(q)_{ij} = \mu_{ij} \exp(-\sigma_{ij}^2 |q|^2/2)$. Each critical wave has one complex amplitude and one K-component eigenvector describing how all cell types participate. For any active set S of waves, the generic cubic equilibrium is $x_S = -G_{SS}^{-1} a_S$, with stability given by one active radial matrix and the invasion rates of the inactive waves. Section 4 derives these formulas and the resonant generalization. Section 5.3 extends them to $\mathbb{S}^1 \times \mathbb{R}^d$, with one discrete orientation harmonic and a continuous d-dimensional cortical wavevector.

V1 is not biologically an E/I two-population system. The Allen VISp/ALM taxonomy has 133 transcriptomic types, and even the deliberately coarse data-driven V1 model of Billeh et al. uses 17 layer-by-class populations. MICrONS resolves cell-to-cell wiring at far finer scale, but no complete physiological gain matrix exists between all transcriptomic types. Therefore this report separates measured type identity and connection probability from calibrated Wilson–Cowan gains. The full published 17-by-17 probability table is supplied with the code; unknown entries are not relabeled as measurements.

The computed coefficients for that example are

$$\alpha = 0.08100 + 0.05273\mathrm{i}, \qquad b = -0.11231 - 0.15236\mathrm{i}, \qquad c = -1.56376 - 0.65446\mathrm{i}. \tag{3}$$

Since $\operatorname{Re} c < \operatorname{Re} b < 0$, the unforced cubic theory selects one traveling wave. At $a = 0.01$ and $\eta_t = \eta_x = 0.04$, the real-coefficient projection predicts a stable standing state. The full complex calculation places every equal-amplitude three-population branch outside stability, but the complete unequal-amplitude quartic reveals a stable exchange-broken locked pair. This distinction is invisible if either the phases or the unequal-amplitude branches are discarded.

The 2025 Drinnenberg et al. preprint identifies a functional split inside deep-L2/3 SST cells: general ensemble-response (SST-GER) neurons integrate broad excitatory activity and differ from local SST and PV cells. Splitting the field into $(E, PV, \mathrm{SST-GER}, \mathrm{SST-local}, VIP)$ retains broad excitation and gives a finite-wave Hopf point at mode 4 and 14.04 Hz, with

$$b_5 = -1.10320 + 0.06808\mathrm{i}, \qquad c_5 = -2.86511 - 0.52395\mathrm{i}. \tag{4}$$

Thus the broader functional partition also selects a traveling wave before forcing. Under the same symmetric forcing it has a stable complex locked standing state of amplitude 0.14949 and phase 172.622° modulo 180°. These are existence calculations, not proof that the same mechanism generates human hallucinations.

On a translation-invariant infinite line, continuous optimization over the wavevector gives a sharper distinction. The three-population traveling wave saturates locally but fails the longitudinal Benjamin–Feir sideband test. The five-population wave passes finite-wave onset, local invasion, and the one-dimensional sideband test. In an isotropic plane or hyperspace, rotational degeneracy still requires an angular or higher-order transverse stability calculation before the pattern can be called fully stable.

2 All amplitude equilibria and their stability

2.1 Reduction to the two real amplitudes

Write

$$A = r_+ \mathrm{e}^{\mathrm{i}\theta_+}, \qquad B = r_- \mathrm{e}^{\mathrm{i}\theta_-}, \qquad r_+, r_- \geq 0, \tag{5}$$

and put $a_r = \operatorname{Re} a$, $b_r = \operatorname{Re} b$, and $c_r = \operatorname{Re} c$. Dividing the first equation in (1) by $\mathrm{e}^{\mathrm{i}\theta_+}$ and separating real and imaginary parts gives

$$\dot{r}_+ = r_+(a_r + b_r r_+^2 + c_r r_-^2), \qquad \dot{\theta}_+ = \operatorname{Im} a + \operatorname{Im} b\, r_+^2 + \operatorname{Im} c\, r_-^2. \tag{6}$$

The second equation gives the reflected pair

$$\dot{r}_- = r_-(a_r + b_r r_-^2 + c_r r_+^2), \qquad \dot{\theta}_- = \operatorname{Im} a + \operatorname{Im} b\, r_-^2 + \operatorname{Im} c\, r_+^2. \tag{7}$$

Thus the imaginary parts change phase and frequency but do not change the amplitude equilibria.

For compactness, set $x = r_+^2$ and $y = r_-^2$. Multiplication of (6) and (7) by $2r_+$ and $2r_-$ gives

$$\dot{x} = 2x(a_r + b_r x + c_r y), \qquad \dot{y} = 2y(a_r + b_r y + c_r x). \tag{8}$$

At an equilibrium, each right-hand side is zero. If both x and y are nonzero, subtraction of the two bracketed equations gives

$$(b_r - c_r)(x - y) = 0. \tag{9}$$

Therefore, when $b_r \neq c_r$, $b_r \neq 0$, and $b_r + c_r \neq 0$, every equilibrium is one of the following. The rest state is

$$(x, y) = (0, 0). \tag{10}$$

The two traveling-wave states are

$$(x, y) \in \left\{ \left(-\frac{a_r}{b_r}, 0 \right), \left(0, -\frac{a_r}{b_r} \right) \right\}. \tag{11}$$

The standing-wave state is

$$(x, y) = \left(-\frac{a_r}{b_r + c_r}, -\frac{a_r}{b_r + c_r} \right). \tag{12}$$

The nonzero state exists exactly when the displayed squared amplitude is positive. The divisions above lose the following degenerate cases. A one-sided state with $x > 0$ exists when

$$a_r + b_r x = 0. \tag{13}$$

Thus $b_r = 0$ gives no one-sided branch when $a_r \neq 0$, but gives every $x > 0$ when $a_r = 0$. When $b_r \neq c_r$, a two-sided state must have $x = y > 0$ and obey

$$a_r + (b_r + c_r)x = 0. \tag{14}$$

Thus $b_r + c_r = 0$ gives no standing branch when $a_r \neq 0$, but leaves every $x = y > 0$ unresolved at cubic order when $a_r = 0$. Finally, if $b_r = c_r \neq 0$, the positive two-sided states form the continuum

$$x + y = -\frac{a_r}{b_r} > 0. \tag{15}$$

If $b_r = c_r = 0$, positive two-sided states exist only when $a_r = 0$, in which case every $x, y > 0$ is a radial equilibrium. Each continuum has a zero radial eigenvalue and requires higher-order terms for selection.

2.2 Jacobian calculation

The radial Jacobian before evaluating an equilibrium is

$$J(r_+, r_-) = \begin{pmatrix} a_r + 3b_r r_+^2 + c_r r_-^2 & 2c_r r_+ r_- \\ 2c_r r_+ r_- & a_r + 3b_r r_-^2 + c_r r_+^2 \end{pmatrix}. \tag{16}$$

At rest, both radial eigenvalues are a_r. Hence rest is stable for $a_r < 0$ and loses stability at $a_r = 0$.

For a traveling wave $(r_+, r_-) = (\rho, 0)$, $\rho^2 = -a_r/b_r$. Substitution into (16) gives

$$J_{\mathrm{TW}} = \begin{pmatrix} -2a_r & 0 \\ 0 & a_r + c_r \rho^2 \end{pmatrix}, \tag{17}$$

and hence

$$\lambda_{\mathrm{radial}} = -2a_r, \qquad \lambda_{\mathrm{opposite}} = a_r \frac{b_r - c_r}{b_r}. \tag{18}$$

The branch can be stable at cubic order only on the supercritical side $a_r > 0$, $b_r < 0$. There the second eigenvalue is negative precisely when

$$c_r < b_r. \tag{19}$$

For a standing wave $r_+ = r_- = \rho$, $\rho^2 = -a_r/(b_r + c_r)$, and

$$J_{\mathrm{SW}} = 2\rho^2 \begin{pmatrix} b_r & c_r \\ c_r & b_r \end{pmatrix}. \tag{20}$$

The symmetric and antisymmetric eigenvectors $(1,1)^T$ and $(1,-1)^T$ give

$$\lambda_{\mathrm{symmetric}} = -2a_r, \qquad \lambda_{\mathrm{imbalance}} = 2\rho^2(b_r - c_r). \tag{21}$$

Thus a supercritical standing wave requires $a_r > 0$, $b_r + c_r < 0$, and is stable to amplitude imbalance precisely when

$$c_r > b_r. \tag{22}$$

The traveling- and standing-wave inequalities are complementary. The two phase eigenvalues are zero because unforced time translation and spatial translation act on $\theta_+ + \theta_-$ and $\theta_+ - \theta_-$. Stability here therefore means orbital stability modulo those symmetries.

The following table summarizes the nondegenerate supercritical cases.

State	Squared amplitude	Supercritical existence	Amplitude stability
Rest	0	all coefficients	$a_r < 0$
Traveling	$-a_r/b_r$	$a_r > 0$, $b_r < 0$	$c_r < b_r$
Standing	$-a_r/(b_r + c_r)$	$a_r > 0$, $b_r + c_r < 0$	$c_r > b_r$

3 Pure 2:1 parametric forcing: $f = 0$

3.1 Why the reduced additive term is exactly absent

Use the weakly nonlinear scaling

$$u = \epsilon u_1 + \epsilon^2 u_2 + \epsilon^3 u_3 + O(\epsilon^4), \qquad T = \epsilon^2 t. \tag{23}$$

Each separated input is now restricted to one of two choices:

Input	Present	Absent
Temporal	$\epsilon^2(P\mathrm{e}^{2\mathrm{i}\omega_0 t} + \overline{P}\mathrm{e}^{-2\mathrm{i}\omega_0 t})$	0
Spatial	$\epsilon^2(S\mathrm{e}^{2\mathrm{i}k_0 x} + \overline{S}\mathrm{e}^{-2\mathrm{i}k_0 x})$	0

Thus $n = 2$ when the temporal input is present and $m = 2$ when the spatial input is present; *absent* may equivalently be denoted by an infinite order. The label 2:1 is a resonance ratio, not an amplitude ratio: the forcing frequency or wavenumber is twice the corresponding critical carrier,

$$\omega_F = 2\omega_0, \qquad k_F = 2k_0. \tag{24}$$

The critical first-order field is

$$u_1 = A(T)\Phi\mathrm{e}^{\mathrm{i}(\omega_0 t + k_0 x)} + B(T)\Phi\mathrm{e}^{\mathrm{i}(\omega_0 t - k_0 x)} + \mathrm{c.c.} \tag{25}$$

The two forcing responses in u_2 are

$$\begin{aligned} p_2 &= (2\mathrm{i}\omega_0 I - M_0(0))^{-1} P, \\ s_2^+ &= -M_0(2k_0)^{-1} S, \qquad s_2^- = -M_0(-2k_0)^{-1}\overline{S}. \end{aligned} \tag{26}$$

At order ϵ^3, the quadratic term $2\mathcal{Q}(u_1, u_2)$ contains

$$\begin{aligned}(2\omega_0, 0) + (-\omega_0, k_0) &= (\omega_0, k_0),\\ (2\omega_0, 0) + (-\omega_0, -k_0) &= (\omega_0, -k_0),\end{aligned} \tag{27}$$

$$\begin{aligned}(0, 2k_0) + (\omega_0, -k_0) &= (\omega_0, k_0),\\ (0, -2k_0) + (\omega_0, k_0) &= (\omega_0, -k_0).\end{aligned} \tag{28}$$

Projection by the adjoint critical eigenvector therefore gives

$$\begin{aligned}\eta_t &= 2\Psi^*\mathcal{Q}(\overline{\Phi}, p_2),\\ \eta_x^+ &= 2\Psi^*\mathcal{Q}(\Phi, s_2^+), \qquad \eta_x^- = 2\Psi^*\mathcal{Q}(\Phi, s_2^-).\end{aligned} \tag{29}$$

Reflection symmetry and a choice of temporal and spatial origins reduce these to $\eta_t, \eta_x \geq 0$ in (2).

The symbol f was not a coefficient of the original unforced amplitude system. It was introduced only as shorthand for a possible amplitude-independent critical residual in a more general forced reduction. If such a third-order residual had critical coefficients R_+ and R_-, the solvability projection would give

$$f_+ = \Psi^* R_+, \qquad f_- = \Psi^* R_-. \tag{30}$$

The calculation below shows that $R_+ = R_- = 0$ for the present exact 2:1-or-absent inputs, so neither f_+ nor f_- appears in (2).

There are two separate reasons that no additive term occurs. First, a temporal forcing harmonic has pair $(2\omega_0, 0)$ and a spatial forcing harmonic has pair $(0, \pm 2k_0)$; neither equals either critical pair $(\omega_0, \pm k_0)$. Second, their first mixed product has order

$$\epsilon^n \epsilon^m = \epsilon^4 \tag{31}$$

and pair $(2\omega_0, \pm 2k_0)$, so it is neither third order nor critical. More strongly, every amplitude-independent monomial formed only from the doubled forcing harmonics has pair

$$(2j\omega_0, 2\ell k_0), \qquad j, \ell \in \mathbb{Z}, \tag{32}$$

which cannot equal $(\omega_0, \pm k_0)$. Hence an analytic nonlinearity cannot generate an additive critical term from exact 2:1 separated forcing at any order. In this report's regime,

$$\boxed{f = 0.} \tag{33}$$

An additive term would require a 1:1 spatio-temporal harmonic, leakage outside the doubled forcing lattice, or explicit translation-breaking heterogeneity; none is assumed here.

If the temporal frequency is $2\omega_0 + \epsilon^2\delta_t$, the parametric terms initially contain $\mathrm{e}^{\mathrm{i}\delta_t T}$. The transformation

$$A = \widetilde{A}\mathrm{e}^{\mathrm{i}\delta_t T/2}, \qquad B = \widetilde{B}\mathrm{e}^{\mathrm{i}\delta_t T/2} \tag{34}$$

makes the system autonomous and replaces a by $a - \mathrm{i}\delta_t/2$. We therefore allow $a, b, c \in \mathbb{C}$ below and use (2) without tildes.

3.2 Exact polar equations and the definition of locking

Write

$$A = r\mathrm{e}^{\mathrm{i}\phi}, \qquad B = \rho\mathrm{e}^{\mathrm{i}\psi}, \qquad r, \rho \geq 0, \tag{35}$$

and separate

$$a + br^2 + c\rho^2 = \alpha_A + \mathrm{i}\beta_A, \qquad a + b\rho^2 + cr^2 = \alpha_B + \mathrm{i}\beta_B. \tag{36}$$

For $r, \rho > 0$, multiplication of (2) by $\mathrm{e}^{-\mathrm{i}\phi}$ and $\mathrm{e}^{-\mathrm{i}\psi}$ gives exactly

$$\dot{r} = r\alpha_A + \eta_t \rho \cos(\phi + \psi) + \eta_x \rho \cos(\phi - \psi), \tag{37}$$

$$r\dot{\phi} = r\beta_A - \eta_t \rho \sin(\phi + \psi) - \eta_x \rho \sin(\phi - \psi), \tag{38}$$

$$\dot{\rho} = \rho\alpha_B + \eta_t r \cos(\phi + \psi) + \eta_x r \cos(\phi - \psi), \tag{39}$$

$$\rho\dot{\psi} = \rho\beta_B - \eta_t r \sin(\phi + \psi) + \eta_x r \sin(\phi - \psi). \tag{40}$$

A phase-locked equilibrium satisfies

$$\dot{r} = \dot{\rho} = \dot{\phi} = \dot{\psi} = 0. \tag{41}$$

The physical carrier still oscillates at frequency ω_0; locking means that both slow amplitudes and both phase offsets relative to the forcing are constant.

Put

$$\Theta = \phi + \psi, \qquad \Delta = \phi - \psi. \tag{42}$$

The radial equations contain

$$C = \eta_t \cos\Theta + \eta_x \cos\Delta \tag{43}$$

and are equivalent to

$$r\alpha_A + \rho C = 0, \qquad \rho\alpha_B + rC = 0. \tag{44}$$

Consequently

$$C = -\frac{r}{\rho}\alpha_A = -\frac{\rho}{r}\alpha_B, \qquad r^2\alpha_A = \rho^2\alpha_B. \tag{45}$$

The phase equations give

$$T := \eta_t \sin\Theta = \frac{1}{2}\left(\frac{r}{\rho}\beta_A + \frac{\rho}{r}\beta_B\right), \tag{46}$$

$$X := \eta_x \sin\Delta = \frac{1}{2}\left(\frac{r}{\rho}\beta_A - \frac{\rho}{r}\beta_B\right). \tag{47}$$

For each $\varepsilon_t, \varepsilon_x \in \{-1, 1\}$, positive amplitudes give a locked state if and only if

$$|T| \le \eta_t, \qquad |X| \le \eta_x, \tag{48}$$

$$-\frac{r}{\rho}\alpha_A = \varepsilon_t\sqrt{\eta_t^2 - T^2} + \varepsilon_x\sqrt{\eta_x^2 - X^2}, \tag{49}$$

together with the last equality in (45). The phases are then

$$\begin{aligned} \Theta &= \operatorname{atan2}\left(T, \varepsilon_t\sqrt{\eta_t^2 - T^2}\right), \\ \Delta &= \operatorname{atan2}\left(X, \varepsilon_x\sqrt{\eta_x^2 - X^2}\right), \end{aligned} \tag{50}$$

$$\phi = \frac{\Theta + \Delta}{2}, \qquad \psi = \frac{\Theta - \Delta}{2}. \tag{51}$$

These conditions are necessary and sufficient, not merely necessary. If $\eta_t = 0$, the first sine condition is replaced by $T = 0$, the term $\eta_t \cos\Theta$ is zero, and Θ is a neutral phase; the first atan2 formula is then omitted. If $\eta_x = 0$, the second sine condition is replaced by $X = 0$, the term $\eta_x \cos\Delta$ is zero, and Δ is neutral; the second atan2 formula is omitted. After assigning arbitrary values to the neutral phases, (51) still reconstructs ϕ and ψ. Thus no undefined atan2(0, 0) is used when an input is absent.

3.3 A factorization that exhausts the positive amplitudes

Let $a_r = \operatorname{Re} a$, $b_r = \operatorname{Re} b$, and $c_r = \operatorname{Re} c$. Expanding the last condition in (45) gives

$$\begin{aligned} 0 &= r^2(a_r + b_r r^2 + c_r \rho^2) - \rho^2(a_r + b_r \rho^2 + c_r r^2) \\ &= (r^2 - \rho^2)\left[a_r + b_r(r^2 + \rho^2)\right]. \end{aligned} \tag{52}$$

Therefore every positive equilibrium satisfies

$$\boxed{r = \rho} \tag{53}$$

or, when $b_r \neq 0$, it satisfies

$$\boxed{r^2 + \rho^2 = -\frac{a_r}{b_r}}. \tag{54}$$

If $b_r = 0$ and $a_r \neq 0$, no unequal-amplitude state exists. If $a_r = b_r = 0$, the factorization is degenerate and the necessary-and-sufficient closure (48)– (49) is used directly.

For the unequal class define only

$$R = r^2 + \rho^2 = -\frac{a_r}{b_r}, \qquad \Pi = r\rho, \qquad D = r^2 - \rho^2 = \pm\sqrt{R^2 - 4\Pi^2}, \tag{55}$$

where $R > 0$ and $0 < \Pi < R/2$. The quantities in the phase closure reduce to

$$C = (b_r - c_r)\Pi, \tag{56}$$

$$T(\Pi) = \frac{a_i R + b_i R^2 + 2(c_i - b_i)\Pi^2}{2\Pi}, \tag{57}$$

$$X(\Pi, D) = \frac{D(a_i + b_i R)}{2\Pi}, \tag{58}$$

where $a_i = \operatorname{Im} a$, $b_i = \operatorname{Im} b$, and $c_i = \operatorname{Im} c$. Hence all unequal-amplitude equilibria are the roots $0 < \Pi < R/2$ of

$$(b_r - c_r)\Pi = \varepsilon_t\sqrt{\eta_t^2 - T(\Pi)^2} + \varepsilon_x\sqrt{\eta_x^2 - X(\Pi, D)^2}, \tag{59}$$

subject to $|T(\Pi)| \leq \eta_t$ and $|X(\Pi, D)| \leq \eta_x$. The two signs of D give the exchanged states. Then

$$r^2 = \frac{R + D}{2}, \qquad \rho^2 = \frac{R - D}{2}, \tag{60}$$

and the phases follow from (50)– (51).

The remaining scalar problem can be reduced to a quartic without losing the sign information. Put $z = \Pi^2$, $d_r = b_r - c_r$, and

$$L = a_i + b_i R, \qquad N(z) = a_i R + b_i R^2 + 2(c_i - b_i)z. \tag{61}$$

The two squared radical arguments, after multiplication by $4z$, are

$$U_0(z) = 4\eta_t^2 z - N(z)^2, \tag{62}$$

$$V_0(z) = 4\eta_x^2 z - (R^2 - 4z)L^2. \tag{63}$$

Thus the unsquared equation has

$$C = d_r\sqrt{z}, \qquad \eta_t^2 - T^2 = \frac{U_0(z)}{4z}, \qquad \eta_x^2 - X^2 = \frac{V_0(z)}{4z}. \tag{64}$$

Eliminating the two square roots gives the explicit quartic equation

$$16d_r^2 z^2 V_0(z) - \left[4d_r^2 z^2 + V_0(z) - U_0(z)\right]^2 = 0. \tag{65}$$

Every unequal equilibrium gives a root of (65). A quartic root is retained exactly when

$$0 < z < \frac{R^2}{4}, \qquad U_0(z) \geq 0, \qquad V_0(z) \geq 0, \tag{66}$$

and it satisfies the original signed equation (59) for at least one pair $(\varepsilon_t, \varepsilon_x)$. This final unsquared test removes every extraneous root introduced by squaring. Consequently there are at most four admissible unequal products Π at generic cubic order, and the two signs of D reconstruct the exchanged states.

3.4 Closed equal-amplitude solutions at every phase

Let $r = \rho > 0$ and first assume $\eta_x > 0$. Then (47) gives $\sin\Delta = 0$, so

$$B = \varepsilon A, \qquad \varepsilon \in \{1, -1\}, \qquad A = r\mathrm{e}^{\mathrm{i}\phi}, \qquad x = r^2. \tag{67}$$

With $s = b + c$ and

$$h_\varepsilon(x) = a + \varepsilon\eta_x + sx, \tag{68}$$

the equilibrium equation is

$$h_\varepsilon(x) + \varepsilon\eta_t \mathrm{e}^{-2\mathrm{i}\phi} = 0. \tag{69}$$

For $\eta_t > 0$, taking the modulus gives

$$|s|^2 x^2 + 2\,\mathrm{Re}[(a + \varepsilon\eta_x)\bar{s}]\, x + |a + \varepsilon\eta_x|^2 - \eta_t^2 = 0. \tag{70}$$

Put

$$D_\varepsilon = \mathrm{Re}[(a + \varepsilon\eta_x)\bar{s}]^2 - |s|^2 \left(|a + \varepsilon\eta_x|^2 - \eta_t^2\right). \tag{71}$$

For $s \neq 0$, all squared amplitudes are

$$x_{\varepsilon,\pm} = \frac{-\,\mathrm{Re}[(a + \varepsilon\eta_x)\bar{s}] \pm \sqrt{D_\varepsilon}}{|s|^2}, \tag{72}$$

retaining exactly the roots with $D_\varepsilon \geq 0$ and $x_{\varepsilon,\pm} > 0$. The corresponding phases are

$$\mathrm{e}^{-2\mathrm{i}\phi} = -\frac{h_\varepsilon(x)}{\varepsilon\eta_t}, \tag{73}$$

$$\phi = -\frac{1}{2}\arg\left[-\frac{h_\varepsilon(x)}{\varepsilon\eta_t}\right] \pmod{\pi}. \tag{74}$$

The two values modulo 2π differ by π and give the sign-related states (A, B) and $(-A, -B)$.

If $s = 0$, the modulus equation is independent of x. There is no positive equal-amplitude branch unless

$$|a + \varepsilon\eta_x| = \eta_t. \tag{75}$$

When this equality holds and $\eta_t > 0$, every $x > 0$ satisfies the cubic-order modulus condition and its phase is given by (73). If $\eta_t = 0$, the undivided condition is $a + \varepsilon\eta_x = 0$ and the common phase is neutral. Each continuum is a degeneracy of the cubic truncation and must be resolved by quintic terms.

For temporal forcing alone, $\eta_x = 0$ and $\eta_t > 0$, the positive equal-amplitude family does not require $B = \pm A$. Instead, for $x = r^2$ it is

$$A = re^{i(\Theta+\Delta)/2}, \qquad B = re^{i(\Theta-\Delta)/2}, \qquad |a + sx| = \eta_t, \qquad e^{-i\Theta} = -\frac{a + sx}{\eta_t}, \tag{76}$$

where Δ is arbitrary. This gives the general locked phase rather than only the special choices $B = \pm A$.

The cases in which one or both inputs are absent follow without a limiting ambiguity:

Forcing present	Equal-amplitude condition	Phase information
Temporal and spatial	$\lvert a + \varepsilon\eta_x + sx\rvert = \eta_t$	$\Delta = 0$ or π; ϕ from (74)
Temporal only	$\lvert a + sx\rvert = \eta_t$	Θ is locked by $a + sx + \eta_t e^{-i\Theta} = 0$; Δ is neutral
Spatial only	$a + \varepsilon\eta_x + sx = 0$	$\Delta = 0$ or π; the common phase is neutral
Neither	$a + sx = 0$ for a stationary complex equilibrium	both group phases are neutral; Section 2 gives the more general relative equilibria

With no forcing, a radial equilibrium whose imaginary frequency correction is nonzero is a relative equilibrium rather than a stationary point in the chosen rotating frame.

3.5 Exceptional one-sided states

Polar coordinates omit $r = 0$ or $\rho = 0$, so return to (2). If $A = 0$ and $B = \rho e^{i\psi} \neq 0$, then

$$\eta_t \overline{B} + \eta_x B = 0, \qquad a + b\rho^2 = 0. \tag{77}$$

If at least one parametric forcing is present, the first equation implies

$$\eta_t = \eta_x > 0, \qquad e^{2i\psi} = -1, \tag{78}$$

and, when $b \neq 0$, the second requires the complex number $-a/b$ to be real and positive, with $\rho^2 = -a/b$. If $b = 0$, it instead requires $a = 0$ and leaves every $\rho > 0$ unresolved at cubic order. Exchanging A and B gives the other one-sided state. If $\eta_t \neq \eta_x$, no nonzero one-sided equilibrium exists. These codimension conditions explain why generic parametric forcing converts unforced traveling waves into two-component locked states. When $\eta_t = \eta_x = 0$, the first condition in (77) disappears and the usual unforced one-sided traveling-wave relative equilibria of Section 2 return.

3.6 Why the equilibrium list is exhaustive

Every point $(A, B) \in \mathbb{C}^2$ belongs to exactly one of the disjoint amplitude classes

$$(|A|, |B|) = (0, 0), \qquad |A||B| = 0 < |A| + |B|, \qquad |A| = |B| > 0, \qquad |A| \neq |B|, \quad |A||B| > 0. \tag{79}$$

The first class is rest. The second is exactly the one-sided calculation (77). In the third class, the phase equations give either (72)– (74), the temporal-only family (76), or one of the absent-input rows in the table above. In the fourth class, (52) forces (54), and the necessary-and-sufficient phase closure reduces it to (59). Hence these formulas omit no complex equilibrium. The exceptional coefficient cases $b_r = 0$ or $a_r = b_r = 0$ are not divided by; they are decided directly by (45) and (49).

3.7 Closed real-coefficient classification

For real a, b, c and real A, B, put

$$q = \eta_t + \eta_x, \qquad s = b + c, \qquad d = b - c. \tag{80}$$

Adding and subtracting the two equilibrium equations gives

$$0 = (A - B)\left[a - q + b(A^2 + B^2) + dAB\right], \tag{81}$$
$$0 = (A + B)\left[a + q + b(A^2 + B^2) - dAB\right]. \tag{82}$$

Therefore the complete real list is:

$$(A, B) = (0, 0),$$
$$(A, B) = (r, r), \qquad r^2 = -\frac{a+q}{s} > 0, \tag{83}$$
$$(A, B) = (r, -r), \qquad r^2 = -\frac{a-q}{s} > 0, \tag{84}$$

plus exchange-broken states. When $bd \neq 0$, the latter satisfy

$$A^2 + B^2 = -\frac{a}{b}, \qquad AB = \frac{q}{d}. \tag{85}$$

Writing $U = -a/b$ and $P = q/d$, they exist exactly when

$$U + 2P \geq 0, \qquad U - 2P \geq 0, \tag{86}$$

and all sign-related and exchanged roots are

$$A = \frac{\sigma\sqrt{U+2P} + \tau\sqrt{U-2P}}{2}, \qquad B = \frac{\sigma\sqrt{U+2P} - \tau\sqrt{U-2P}}{2}, \quad \sigma, \tau \in \{-1, 1\}. \tag{87}$$

Equalities in (86) merge this family with an equal or anti-equal branch. If $b = 0$, $d = 0$, or $s = 0$, the undivided factorization (81)–(82) gives the corresponding degenerate cases without losing a branch.

3.8 Exact stability: rest and every equal-amplitude branch

At the rest state, exchange-symmetric and exchange-antisymmetric perturbations satisfy

$$\dot{z}_\kappa = (a + \kappa\eta_x) z_\kappa + \kappa\eta_t \overline{z}_\kappa, \qquad \kappa \in \{1, -1\}. \tag{88}$$

Their characteristic polynomials and eigenvalues are

$$0 = \lambda^2 - 2\,\mathrm{Re}(a + \kappa\eta_x)\lambda + |a + \kappa\eta_x|^2 - \eta_t^2, \tag{89}$$
$$\lambda_{\kappa,\pm} = \mathrm{Re}\, a + \kappa\eta_x \pm \sqrt{\eta_t^2 - (\mathrm{Im}\, a)^2}. \tag{90}$$

The square root is imaginary when its argument is negative. The rest state is asymptotically stable if and only if, for both $\kappa = \pm 1$,

$$\mathrm{Re}(a + \kappa\eta_x) < 0, \qquad |a + \kappa\eta_x|^2 > \eta_t^2. \tag{91}$$

Now let $B = \varepsilon A$, $A = re^{i\phi}$, and $x = r^2$. The full 4×4 linearization splits exactly into the two exchange sectors $\delta B = \kappa \delta A$. In sector κ it is the scalar conjugate-linear equation

$$\delta \dot{A} = P_\kappa \delta A + Q_\kappa \delta \overline{A}, \tag{92}$$

where

$$P_\kappa = a + [2b + c + \kappa\varepsilon c]\, x + \kappa \eta_x, \tag{93}$$

$$Q_\kappa = x e^{2i\phi}(b + \kappa\varepsilon c) + \kappa \eta_t. \tag{94}$$

The two eigenvalues in that sector obey

$$\lambda^2 - 2\,\mathrm{Re}\, P_\kappa\, \lambda + |P_\kappa|^2 - |Q_\kappa|^2 = 0 \tag{95}$$

and therefore

$$\lambda_{\kappa,\pm} = \mathrm{Re}\, P_\kappa \pm \sqrt{|Q_\kappa|^2 - (\mathrm{Im}\, P_\kappa)^2}. \tag{96}$$

Thus an equal-amplitude locked state is asymptotically stable if and only if, for each $\kappa = \pm 1$,

$$\boxed{\mathrm{Re}\, P_\kappa < 0, \qquad |P_\kappa|^2 > |Q_\kappa|^2.} \tag{97}$$

If one forcing is absent, the continuous symmetry it leaves unbroken produces the expected zero eigenvalue; stability is then orbital after removing that neutral group direction.

3.9 Exact stability at every remaining complex root

Let the right-hand side of the first equation in (2) be F_A. Its Wirtinger derivatives are

$$\begin{aligned} \partial_A F_A &= a + 2b|A|^2 + c|B|^2, & \partial_{\overline{A}} F_A &= bA^2, \\ \partial_B F_A &= cA\overline{B} + \eta_x, & \partial_{\overline{B}} F_A &= cAB + \eta_t. \end{aligned} \tag{98}$$

The derivatives of F_B follow by exchanging A and B. The real matrix of $z \mapsto Pz + Q\overline{z}$ is

$$\mathcal{R}(P, Q) = \begin{pmatrix} \mathrm{Re}(P+Q) & -\mathrm{Im}(P-Q) \\ \mathrm{Im}(P+Q) & \mathrm{Re}(P-Q) \end{pmatrix}. \tag{99}$$

Using (99) on the four derivative pairs in (98) and its exchanged copy produces the exact real 4×4 Jacobian

$$J = \begin{pmatrix} \mathcal{R}(a + 2b|A|^2 + c|B|^2, bA^2) & \mathcal{R}(cA\overline{B} + \eta_x, cAB + \eta_t) \\ \mathcal{R}(cB\overline{A} + \eta_x, cAB + \eta_t) & \mathcal{R}(a + 2b|B|^2 + c|A|^2, bB^2) \end{pmatrix}. \tag{100}$$

This formula applies at any root reconstructed above, including the codimension-one-sided states.

For a coefficient-only stability test, write

$$\det(\lambda I - J) = \lambda^4 + h_1\lambda^3 + h_2\lambda^2 + h_3\lambda + h_4, \tag{101}$$

where no eigenvalue calculation is required:

$$\begin{aligned} h_1 &= -\,\mathrm{tr}\, J, \\ h_2 &= \frac{1}{2}\left[(\mathrm{tr}\, J)^2 - \mathrm{tr}(J^2)\right], \\ h_3 &= -\frac{1}{6}\left[(\mathrm{tr}\, J)^3 - 3\,\mathrm{tr}\, J\ \mathrm{tr}(J^2) + 2\,\mathrm{tr}(J^3)\right], \\ h_4 &= \det J. \end{aligned} \tag{102}$$

The root is asymptotically stable if and only if the quartic Routh–Hurwitz inequalities hold:

$$h_1, h_2, h_3, h_4 > 0, \qquad h_1 h_2 > h_3, \qquad h_1 h_2 h_3 > h_3^2 + h_1^2 h_4. \tag{103}$$

Equations (59) and (103) therefore give a complete existence-and-stability algorithm for every unequal complex phase-locked state. Applying the same Jacobian and Hurwitz test to (77) gives the exact stability of every exceptional one-sided state.

3.10 Application to the computed three- and five-population coefficients

For the real-coefficient projection, take $a = 0.01$ and $\eta_t = \eta_x = 0.04$. The real parts of (3) give

$$(A, B) = (0, 0), \qquad (A, B) = (r, r), \qquad (A, B) = (-r, -r), \qquad r = 0.23173. \tag{104}$$

The rest state is unstable. Each nonzero state has eigenvalues

$$(-0.18000, -0.00412, -0.08000, -0.08000) \tag{105}$$

and is asymptotically stable within this real-coefficient projection. These real states are not equilibria of the full complex-coefficient normal form.

At zero detuning, retain the complete complex values of b, c and apply (72)– (74). The exact sector formula (96) gives:

Model	ε	r	ϕ (degrees)	$\max \mathrm{Re}\,\lambda$	Stability
Three-population	$+1$	0.20564	150.732	0.00100	unstable
Three-population	$+1$	0.07843	93.563	0.07839	unstable
Three-population	-1	0.07617	86.639	0.08000	unstable
Five-population	$+1$	0.14949	172.622	-0.07861	stable
Five-population	$+1$	0.05024	90.824	0.07991	unstable
Five-population	-1	0.05016	89.178	0.08000	unstable

Thus all equal-amplitude branches of the full three-population model are unstable at this forcing, although the large branch is only weakly unstable. The five-population model has the stable complex locked state

$$(A, B) = \left(0.14949 \mathrm{e}^{\mathrm{i}\,172.622^\circ}, 0.14949 \mathrm{e}^{\mathrm{i}\,172.622^\circ}\right) \tag{106}$$

and its sign-related partner.

The equal-amplitude table is not the complete numerical list. For both models the one-sided condition fails because $-a/b$ is not real. Solving the quartic (65) and applying the unsquared filter (59) gives the following representatives:

Model	r	ρ	ϕ (degrees)	ψ (degrees)	$\max \mathrm{Re}\,\lambda$	Stability
Three-population	0.06978	0.29011	-24.559	-66.184	0.01390	unstable
Three-population	0.15984	0.25197	-26.250	-35.442	-0.00181	stable
Five-population	0.00077	0.09520	-16.561	89.904	0.07233	unstable

Each row has an exchanged partner obtained by interchanging (r, ϕ) with (ρ, ψ), and each of those has a sign-related partner obtained by adding 180° to both phases. Substitution into the full four-real-dimensional system gives residuals below 2×10^{-17} before rounding the displayed values.

Therefore the complete calculation changes the activation-pattern conclusion for the three-population coefficients: although every equal-amplitude locked branch is unstable, there is a stable exchange-broken phase-locked pair. For the five-population coefficients, the stable state remains the equal-amplitude branch (106); its unequal pair is unstable.

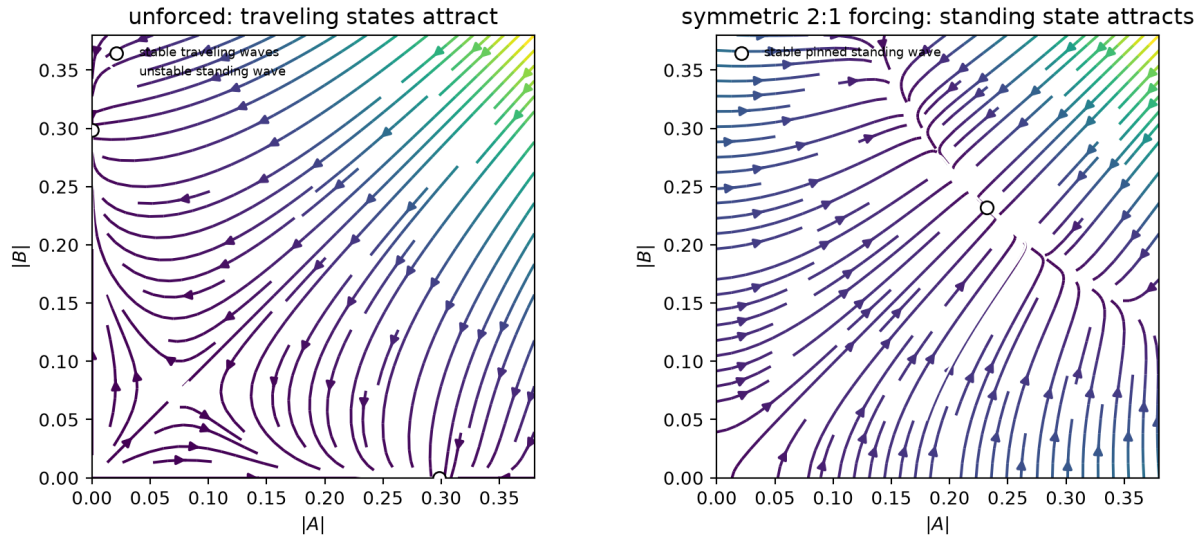


Figure 1: Amplitude phase diagrams using the computed real cubic coefficients. White circles are stable states and the cross is the unstable unforced standing wave.

4 Closed K-population Gaussian and wave framework

4.1 The two connectivity matrices

Let $u = (u_1, \dots, u_K)^T$ collect the activities of K cell populations. The signed Gaussian mass and width matrices are

$$\boldsymbol{\mu} = (\mu_{ij}) \in \mathbb{R}^{K\times K}, \qquad \Sigma = (\sigma_{ij}) \in (0,\infty)^{K\times K}, \tag{107}$$

where row i is the target and column j is the source. Normalize the Gaussian to unit mass and set

$$G_\sigma(x) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{|x|^2}{2\sigma^2}\right), \qquad J_{ij}(x) = \mu_{ij} G_{\sigma_{ij}}(x). \tag{108}$$

Thus $\mu_{ij} = \int J_{ij}$ is the total signed connection gain, not its peak. If data give a Gaussian peak p_{ij}, then $\mu_{ij} = p_{ij}(2\pi)^{d/2}\sigma_{ij}^d$. The field equation is

$$\mathsf{T}\partial_t u = -u + F(J * u + I), \qquad \mathsf{T} = \operatorname{diag}(\tau_1, \dots, \tau_K), \tag{109}$$

with F acting componentwise. On $\mathbb{R}^d$, or on a torus using wrapped Gaussians, its Fourier connectivity matrix is the closed expression

$$\widehat{J}(q) = \boldsymbol{\mu} \odot \exp\left[-\frac{|q|^2}{2}\Sigma^{\odot 2}\right], \qquad \widehat{J}(q)_{ij} = \mu_{ij} \mathrm{e}^{-\sigma_{ij}^2|q|^2/2}, \tag{110}$$

where the exponential and powers are entrywise. This is the only place where the Gaussian assumption enters the spectral calculation.

4.2 Homogeneous equilibria and exact field stability

Every homogeneous equilibrium u^* is a root of the K equations

$$u^* = F(\boldsymbol{\mu} u^* + I). \tag{111}$$

This is an exact finite-dimensional reduction because every kernel has unit mass. No formula by radicals exists for an arbitrary choice of K nonlinear functions F_i. On an affine or threshold-linear branch $F(h) = Sh + c$, however, the equilibrium is explicitly

$$u^* = (I - S\boldsymbol{\mu})^{-1}(SI + c), \tag{112}$$

provided $I - S\boldsymbol{\mu}$ is nonsingular and the result satisfies the inequalities defining that branch. For an invertible sigmoid, (111) is equivalently $F^{-1}(u^*) - \boldsymbol{\mu}u^* = I$ and is solved as a K-variable root.

Let $D_r = \mathrm{diag}(F_i^{(r)}(h_i^*))$, where $h^* = \boldsymbol{\mu}u^* + I$. A perturbation $v\mathrm{e}^{\lambda t + \mathrm{i}q\cdot x}$ satisfies

$$\lambda v = M(q)v, \qquad M(q) = \mathsf{T}^{-1}\left[-I + D_1\widehat{J}(q)\right]. \tag{113}$$

Consequently the equilibrium is linearly asymptotically stable precisely when

$$\sup_{q\in\mathcal{Q}} \max_{1\le s\le K} \mathrm{Re}\,\lambda_s(M(q)) < 0, \tag{114}$$

where $\mathcal{Q}$ is continuous on $\mathbb{R}^d$ and is the reciprocal lattice on a periodic domain. A zero of the left side at $q = 0$ is a homogeneous instability; a zero first reached at $|q| > 0$ is a pattern-forming instability. This criterion includes every cell type and every spatial wave, without reducing $\boldsymbol{\mu}$ to an effective E/I difference.

For modal vectors v, w at wavevectors q_1, q_2, write $\ell_q(v) = \widehat{J}(q)v$. Taylor expansion of F gives the symmetric quadratic and cubic maps

$$\mathcal{Q}_{q_1,q_2}(v,w) = \frac{1}{2}\mathsf{T}^{-1}D_2\left[\ell_{q_1}(v)\odot\ell_{q_2}(w)\right], \tag{115}$$

$$\mathcal{C}_{q_1,q_2,q_3}(u,v,w) = \frac{1}{6}\mathsf{T}^{-1}D_3\left[\ell_{q_1}(u)\odot\ell_{q_2}(v)\odot\ell_{q_3}(w)\right]. \tag{116}$$

Equations (110), (113), (115), and (116) express the complete cubic local dynamics in terms of $\boldsymbol{\mu}$, Σ, T, and the derivatives of the K firing-rate functions.

4.3 Cell types carried by critical waves

Suppose m critical wave types have wavevectors q_α, frequencies ω_α, and right and adjoint cell-type vectors satisfying

$$M(q_\alpha)\Phi_\alpha = \mathrm{i}\omega_\alpha\Phi_\alpha, \qquad M(q_\alpha)^*\Psi_\alpha = -\mathrm{i}\omega_\alpha\Psi_\alpha, \qquad \Psi_\alpha^*\Phi_\alpha = 1. \tag{117}$$

The leading wave superposition is

$$u_i(x,t) = u_i^* + 2\varepsilon\,\mathrm{Re}\sum_{\alpha=1}^{m} A_\alpha(T)\Phi_{i\alpha}\mathrm{e}^{\mathrm{i}(q_\alpha\cdot x + \omega_\alpha t)} + O(\varepsilon^2), \qquad T = \varepsilon^2 t. \tag{118}$$

There is one complex dynamical variable A_α per wave, not per cell type. The entry $\Phi_{i\alpha}$ gives the amplitude and phase with which cell type i participates in wave α. Thus the K-cell biology is retained inside each wave rather than replaced by Km redundant amplitude variables.

For completeness, the general cubic projection can be written without listing every conjugate case. Give every critical wave a signed copy $r = (\alpha, s)$, $s = \pm 1$, and define

$$z_{\alpha,+} = A_\alpha, \quad z_{\alpha,-} = \overline{A}_\alpha, \quad q_{\alpha,s} = sq_\alpha, \quad \omega_{\alpha,s} = s\omega_\alpha, \quad \phi_{\alpha,+} = \Phi_\alpha, \quad \phi_{\alpha,-} = \overline{\Phi}_\alpha. \tag{119}$$

For a noncritical pair (r, s), its slaved second-order cell-type response is

$$H_{rs} = [\mathrm{i}(\omega_r + \omega_s)I - M(q_r + q_s)]^{-1}\, \mathcal{Q}_{q_r,q_s}(\phi_r, \phi_s). \tag{120}$$

The inverse is taken only off the critical eigenspace. Summing over ordered signed indices, the amplitude equation through cubic order is

$$\begin{aligned}\dot{A}_\alpha = \lambda_\alpha A_\alpha &+ \sum_{\substack{q_r+q_s=q_\alpha \\ \omega_r+\omega_s=\omega_\alpha}} \Psi^*_\alpha \mathcal{Q}_{q_r,q_s}(\phi_r, \phi_s) z_r z_s \\ &+ \sum_{\substack{q_r+q_s+q_t=q_\alpha \\ \omega_r+\omega_s+\omega_t=\omega_\alpha}} \Psi^*_\alpha [\mathcal{C}_{q_r,q_s,q_t}(\phi_r, \phi_s, \phi_t) + 2\mathcal{Q}_{q_r,q_s+q_t}(\phi_r, H_{st})]\, z_r z_s z_t + \sum_{\beta=1}^{m} \left(\eta^x_{\alpha\beta} A_\beta + \eta^t_{\alpha\beta} \overline{A}_\beta\right).\end{aligned} \tag{121}$$

Here λ_α is the projected parameter detuning. $\eta^x_{\alpha\beta}$ is nonzero only when an order-2 spatial forcing harmonic scatters wave β into wave α, while $\eta^t_{\alpha\beta}$ is nonzero only when an order-2 temporal forcing harmonic scatters the conjugate of wave β into wave α. They are the multi-wave analogues of η_x and η_t in (2). There is no additive vector in the order-2 or absent regime by (32). The two equalities under each nonlinear sum are conservation of wavevector and temporal frequency. If a quadratic pair is itself critical, it is retained in the first sum and is not inverted in (120). Equation (121) is the desired wave-based dynamical system for arbitrary K and any finite collection of interacting waves.

4.4 Closed equilibria and stability of a wave superposition

Away from special quadratic and phase resonances, translation and time-shift symmetry reduce (121) to

$$\dot{A}_\alpha = A_\alpha \left(\lambda_\alpha + \sum_{\beta=1}^{m} g_{\alpha\beta} |A_\beta|^2\right), \qquad \alpha = 1, \ldots, m. \tag{122}$$

Put $x_\alpha = |A_\alpha|^2$, $a_\alpha = \operatorname{Re} \lambda_\alpha$, and $G_{\alpha\beta} = \operatorname{Re} g_{\alpha\beta}$. Then

$$\dot{x}_\alpha = 2x_\alpha \left(a_\alpha + \sum_{\beta=1}^{m} G_{\alpha\beta} x_\beta\right). \tag{123}$$

Every equilibrium is indexed by its active wave set S. If G_{SS} is nonsingular, its closed form is

$$x_S = -G_{SS}^{-1} a_S > 0, \qquad x_\alpha = 0 \quad (\alpha \notin S). \tag{124}$$

Thus all generic cubic wave equilibria are obtained by checking the at most 2^m supports; negative components are discarded. At such a state the active radial Jacobian and the inactive invasion rates are

$$J_S = 2\,\mathrm{diag}(x_S) G_{SS}, \qquad \nu_\alpha = 2\,(a_\alpha + G_{\alpha S} x_S) \quad (\alpha \notin S). \tag{125}$$

For one active wave α, this reads $x_\alpha = -a_\alpha / G_{\alpha\alpha}$. For two active waves α, β, put $\Delta = G_{\alpha\alpha} G_{\beta\beta} - G_{\alpha\beta} G_{\beta\alpha}$. Their explicit intensities are

$$x_\alpha = \frac{G_{\alpha\beta} a_\beta - G_{\beta\beta} a_\alpha}{\Delta}, \qquad x_\beta = \frac{G_{\beta\alpha} a_\alpha - G_{\alpha\alpha} a_\beta}{\Delta}. \tag{126}$$

These are retained only when both are positive; their radial and invasion stability then follows directly from (125). The wave superposition is asymptotically stable modulo phase symmetries if every eigenvalue of J_S has negative real part and every $\nu_\alpha < 0$. Its slow phase velocities are

$$\dot{\theta}_\alpha = \operatorname{Im} \lambda_\alpha + \sum_\beta \operatorname{Im} g_{\alpha\beta} x_\beta. \tag{127}$$

They give nonlinear frequency corrections; forcing or resonant monomials can lock them and remove the neutral phase directions. When such terms are present, equilibria are roots of the explicit polynomial system (121), and stability is decided by its real $2m \times 2m$ Jacobian. There is no universal radical formula for that fully resonant problem, but the dynamical system and Jacobian remain closed.

4.5 The counter-propagating pair as a special case

At a single critical q_0, write $\Phi = \Phi_1$, $\Psi = \Psi_1$ and

$$R(\Omega, q) = (\mathrm{i}\Omega I - M(q))^{-1}. \tag{128}$$

The second-order fields required for the self coefficient are

$$\begin{aligned} H_{11}^0 &= 2R(0,0)\mathcal{Q}_{q_0,-q_0}(\Phi, \overline{\Phi}), \\ H_{20}^{2q_0} &= R(2\omega_0, 2q_0)\mathcal{Q}_{q_0,q_0}(\Phi, \Phi), \end{aligned} \tag{129}$$

and the self coefficient is

$$\begin{aligned} b = \Psi^* \big[&2\mathcal{Q}_{q_0,0}(\Phi, H_{11}^0) + 2\mathcal{Q}_{-q_0,2q_0}(\overline{\Phi}, H_{20}^{2q_0}) \\ &+ 3\mathcal{C}_{q_0,q_0,-q_0}(\Phi, \Phi, \overline{\Phi}) \big]. \end{aligned} \tag{130}$$

For the counter-propagating coefficient, define

$$\begin{aligned} H_{20}^0 &= 2R(2\omega_0, 0)\mathcal{Q}_{q_0,-q_0}(\Phi, \Phi), \\ H_{11}^{2q_0} &= 2R(0, 2q_0)\mathcal{Q}_{q_0,q_0}(\Phi, \overline{\Phi}), \end{aligned} \tag{131}$$

and

$$\begin{aligned} c = \Psi^* \big[&2\mathcal{Q}_{q_0,0}(\Phi, H_{11}^0) + 2\mathcal{Q}_{q_0,0}(\overline{\Phi}, H_{20}^0) + 2\mathcal{Q}_{-q_0,2q_0}(\Phi, H_{11}^{2q_0}) \\ &+ 6\mathcal{C}_{q_0,-q_0,q_0}(\Phi, \Phi, \overline{\Phi}) \big]. \end{aligned} \tag{132}$$

If a scalar bifurcation parameter changes M by M_1, then

$$\alpha = \Psi^* M_1(q_0)\Phi, \qquad a = \mu\alpha. \tag{133}$$

This recovers (1): $g_{11} = g_{22} = b$ and $g_{12} = g_{21} = c$. Every coefficient is obtained from K-vectors, $K \times K$ matrix solves, and adjoint projections.

5 Rigorous translation-invariant domain extensions

5.1 Extension 1: the infinite line

First let $\Omega = \mathbb{R}$. The Fourier transform is continuous rather than a Fourier series. Equations (110) and (113) become

$$\widehat{J}(q)_{ij} = \mu_{ij} \mathrm{e}^{-\sigma_{ij}^2 q^2/2}, \qquad q \in \mathbb{R}, \tag{134}$$

$$\partial_t \widehat{v}(q,t) = M(q)\widehat{v}(q,t), \qquad M(q) = \mathsf{T}^{-1}\left[-I + D_1 \widehat{J}(q)\right]. \tag{135}$$

Equivalently, the linearized operator is the direct integral

$$\mathcal{F}\mathcal{L}\mathcal{F}^{-1} = \int_{\mathbb{R}}^{\oplus} M(q)\, dq, \qquad \operatorname{spec}(\mathcal{L}) = \overline{\bigcup_{q\in\mathbb{R}} \operatorname{spec}(M(q))}. \tag{136}$$

Plancherel's theorem gives

$$\|v(t)\|_{L^2}^2 = \int_{\mathbb{R}} \|\mathrm{e}^{tM(q)}\widehat{v}(q,0)\|_{\mathbb{C}^K}^2\, dq. \tag{137}$$

Hence a homogeneous equilibrium is exponentially stable when there are $C, \gamma > 0$ such that

$$\|\mathrm{e}^{tM(q)}\| \le C\mathrm{e}^{-\gamma t} \quad (q \in \mathbb{R}, t \ge 0), \tag{138}$$

For the Gaussian symbol this is equivalent to the spectral-gap condition

$$\sup_{q\in\mathbb{R}} \max_{s} \operatorname{Re}\lambda_s(M(q)) < 0. \tag{139}$$

To see the nontrivial implication, use $M(q) \to -\mathsf{T}^{-1}$ as $|q| \to \infty$, so the matrix family is continuous on the compactification $\mathbb{R}\cup\{\infty\}$. If the left side of (139) is -2γ, choose one contour Γ enclosing all spectra in $\{\operatorname{Re} z \le -\gamma\}$. Then

$$\mathrm{e}^{tM(q)} = \frac{1}{2\pi\mathrm{i}} \int_\Gamma \mathrm{e}^{tz}(zI - M(q))^{-1}\, dz, \qquad \sup_q \|\mathrm{e}^{tM(q)}\| \le \frac{|\Gamma|}{2\pi} \sup_{q,z\in\Gamma} \|(zI - M(q))^{-1}\| \mathrm{e}^{-\gamma t}. \tag{140}$$

The reverse implication follows by applying (138) to an eigenvector of $M(q)$. At threshold, a carrier satisfies

$$M(q_0)\Phi = \mathrm{i}\omega_0 \Phi. \tag{141}$$

It is a generalized Fourier mode rather than an $L^2(\mathbb{R})$ eigenfunction, because $\Phi \mathrm{e}^{\mathrm{i}q_0 x}$ has constant modulus. Consequently the whole-line center spectrum is not a finite-dimensional eigenspace. The two carrier amplitudes describe spatially uniform envelopes only; localized perturbations require the modulation equation derived in Extension 2.

The extension used from this point onward is translation invariant. Thus the spatial domain is $\mathbb{R}^d$, not a bounded subset, the coupling is a true convolution, and every nonlinear interaction retains exact wavevector conservation. A bounded truncation is a different model because its edges break the translation group and pin the phase of a pattern.

5.2 Extension 2: non-periodic space in arbitrary dimension

Let $x \in \mathbb{R}^d$ and allow a positive-definite covariance $C_{ij} \in \mathbb{R}^{d\times d}$ for each ordered cell-type pair:

$$G_{C_{ij}}(x) = \frac{\exp[-\frac{1}{2}x^T C_{ij}^{-1} x]}{(2\pi)^{d/2}\det(C_{ij})^{1/2}}, \qquad J_{ij}(x) = \mu_{ij} G_{C_{ij}}(x), \tag{142}$$

$$\widehat{J}(q)_{ij} = \mu_{ij}\exp[-\tfrac{1}{2}q^T C_{ij} q], \qquad q \in \mathbb{R}^d. \tag{143}$$

The scalar-width model is recovered by $C_{ij} = \sigma_{ij}^2 I_d$. The full stability criterion is

$$\sup_{q\in\mathbb{R}^d} \max_s \operatorname{Re}\lambda_s\Big(\mathsf{T}^{-1}[-I + D_1\widehat{J}(q)]\Big) < 0. \tag{144}$$

5.2.1 Exact linear condition for a hallucination-like pattern

In this report, a hallucination-like state means a self-sustained, nonuniform cortical solution of the field model. This mathematical definition does not by itself establish a subjective human hallucination. Let δ move the homogeneous equilibrium through onset and define

$$M(q,\delta) = \mathsf{T}^{-1}\left[-I + D_1(\delta)\widehat{J}(q,\delta)\right], \qquad s(q,\delta) = \max_{1\le j\le K} \operatorname{Re}\lambda_j(M(q,\delta)). \tag{145}$$

The homogeneous state is exponentially stable exactly when

$$\sup_{q\in\mathbb{R}^d} s(q,\delta) < 0. \tag{146}$$

A finite-wave stationary or oscillatory instability occurs at $\delta = 0$ if there are $q_0 \neq 0$, a simple eigenvalue branch λ_*, and $\omega_0 \geq 0$ such that

$$\lambda_*(q_0,0) = \mathrm{i}\omega_0, \qquad s(0,0) < 0, \qquad \sup_{q\in\mathbb{R}^d} s(q,0) = 0. \tag{147}$$

The crossing is nondegenerate at an isolated anisotropic maximum when

$$\partial_\delta \operatorname{Re}\lambda_*(q_0,0) > 0, \qquad \nabla_q \operatorname{Re}\lambda_*(q_0,0) = 0, \qquad \xi^T\nabla_q^2 \operatorname{Re}\lambda_*(q_0,0)\xi < 0 \quad (\xi \neq 0). \tag{148}$$

For isotropic kernels, write $r = |q|$. The critical set is the sphere $|q| = q_0$, so the tangential Hessian directions are neutral by rotation symmetry and the correct radial condition is

$$\partial_r \operatorname{Re}\lambda_*(q_0,0) = 0, \qquad \partial_r^2 \operatorname{Re}\lambda_*(q_0,0) < 0. \tag{149}$$

The case $\omega_0 = 0$ is a stationary finite-wave instability. The case $\omega_0 > 0$ is a finite-wave Hopf instability and produces oscillating or traveling cortical patterns. Immediately above onset, the necessary linear pattern condition is therefore

$$s(0,\delta) < 0, \qquad \sup_{q\neq 0} s(q,\delta) > 0. \tag{150}$$

This says that a nonzero spatial scale grows while the spatially homogeneous mode remains damped. It is not yet sufficient, because nonlinear growth must saturate and the resulting pattern must resist all long-wave modulations.

Assume the critical eigenvalue bands are simple and smooth, all other bands remain uniformly separated from the imaginary axis, and the first crossing occurs at a finite set $q_1, \dots, q_m$. The cell-resolved expansion is then

$$u_i(x,t) = u_i^* + 2\varepsilon \,\mathrm{Re} \sum_{\alpha=1}^{m} A_\alpha(X,T)\Phi_{i\alpha} \mathrm{e}^{\mathrm{i}(q_\alpha \cdot x + \omega_\alpha t)} + O(\varepsilon^2), \quad X = \varepsilon x,\ T = \varepsilon^2 t. \tag{151}$$

Let $\lambda_\alpha(q,\delta)$ be the simple eigenvalue branch and put $\delta = \varepsilon^2\mu$. Its Taylor expansion is

$$\begin{aligned} \lambda_\alpha(q_\alpha + \varepsilon\kappa, \varepsilon^2\mu) = \mathrm{i}\omega_\alpha + \varepsilon \nabla_q \lambda_\alpha(q_\alpha, 0)\cdot\kappa \\ + \varepsilon^2 \left[\alpha_\alpha \mu + \frac{1}{2}\kappa^T H_\alpha \kappa\right] + O(\varepsilon^3), \end{aligned} \tag{152}$$

where

$$\alpha_\alpha = \partial_\delta \lambda_\alpha(q_\alpha, 0), \qquad H_\alpha = \nabla_q^2 \lambda_\alpha(q_\alpha, 0). \tag{153}$$

After removing the order-ε transport by a comoving coordinate, the envelope equation whose Fourier symbol matches (152) is

$$\partial_T A_\alpha = \alpha_\alpha \mu A_\alpha - \frac{1}{2}\sum_{r,s=1}^{d} (H_\alpha)_{rs} \partial_{X_r X_s} A_\alpha + A_\alpha \sum_{\beta=1}^{m} g_{\alpha\beta} |A_\beta|^2. \tag{154}$$

Indeed, substituting $A_\alpha = \mathrm{e}^{\mathrm{i}\kappa\cdot X + \Lambda T}$ into the linear part gives

$$\Lambda = \alpha_\alpha \mu + \frac{1}{2}\kappa^T H_\alpha \kappa, \tag{155}$$

which is exactly the order-ε^2 term in (152).

The finite-dimensional wave equations describe only envelopes that are constant in X. To test a constant equilibrium against every localized sideband, denote the local right-hand side of (154) by $\mathcal{N}(A, \overline{A})$ after removing the derivative term. Define its doubled real-linear Jacobian by

$$\mathbb{J}_0 = D_{(A,\overline{A})} \begin{pmatrix} \mathcal{N} \\ \overline{\mathcal{N}} \end{pmatrix} \Big|_{A=A^*}. \tag{156}$$

For a perturbation proportional to $\mathrm{e}^{\mathrm{i}\kappa\cdot X + \Lambda T}$, put

$$d_\alpha(\kappa) = \frac{1}{2}\kappa^T H_\alpha \kappa. \tag{157}$$

The exact $2m \times 2m$ sideband matrix is

$$\mathbb{J}(\kappa) = \mathbb{J}_0 + \mathrm{diag}\Big(d_1(\kappa), \dots, d_m(\kappa), \overline{d_1(\kappa)}, \dots, \overline{d_m(\kappa)}\Big). \tag{158}$$

Therefore orbital modulational stability requires

$$\max_{\lambda \in \mathrm{spec}(\mathbb{J}(\kappa))} \mathrm{Re}\,\lambda < 0 \quad (\kappa \neq 0), \qquad \mathrm{spec}(\mathbb{J}(0)) \subset \{z : \mathrm{Re}\, z < 0\} \cup \{0\}. \tag{159}$$

The kernel at $\kappa = 0$ must consist exactly of the phase and translation directions generated by symmetry. If $\lambda_\ell^{\mathrm{sym}}(\kappa)$ are the spectral branches issuing from that kernel, diffusive decay requires constants $c, \kappa_0 > 0$ such that

$$\mathrm{Re}\,\lambda_\ell^{\mathrm{sym}}(\kappa) \le -c|\kappa|^2 \qquad (0 < |\kappa| < \kappa_0). \tag{160}$$

The Jacobian test in (125) is precisely the $\kappa = 0$ part and is not, by itself, a sideband-stability test.

5.2.2 Necessary and sufficient conditions within the amplitude regime

Let S be the active set of a finite planform and let $x_S = -G_{SS}^{-1} a_S$. Within the validity of the cubic modulation equations, existence and homogeneous-amplitude stability are equivalent to

$$x_S > 0, \qquad \max_{\lambda \in \mathrm{spec}(J_S)} \mathrm{Re}\,\lambda < 0, \qquad \nu_\alpha < 0 \quad (\alpha \notin S). \tag{161}$$

Here J_S and ν_α are given exactly by (125). The planform is a stable hallucination-like state precisely when (161), (159), and (160) all hold. A spatially varying prescribed input is absent from this criterion: it would select an origin and break translation invariance. Spatially uniform parameter changes or temporal forcing can preserve the spatial translation group.

For isotropic kernels, $M(q)$ depends only on $|q|$ and the critical set is the sphere $|q| = q_0$. It is then not a finite-dimensional center eigenspace. The correct leading superposition is

$$u_i(x,t) = u_i^* + 2\varepsilon\,\mathrm{Re} \int_{\mathbb{S}^{d-1}} A(\xi, X, T)\Phi_i(\xi) \mathrm{e}^{\mathrm{i}(q_0 \xi \cdot x + \omega_0 t)}\, d\xi + O(\varepsilon^2). \tag{162}$$

Its cubic interactions obey

$$q_0(\xi_1 + \xi_2 - \xi_3) = q_0 \xi, \qquad \omega_1 + \omega_2 - \omega_3 = \omega, \tag{163}$$

and their coefficients are still obtained from (115)–(120). Choosing a finite planform $\{\xi_1, \ldots, \xi_m\}$ and restricting to spatially constant envelopes reduces (162) to the finite wave system (122). Its equilibria and homogeneous-amplitude stability are exactly (124)– (125); sidebands are governed by (158).

For two wavevectors separated by ϑ,

$$|q_j + q_k| = 2q_0 \cos(\vartheta/2), \qquad |q_j - q_k| = 2q_0 \sin(\vartheta/2), \tag{164}$$

so $g_{jk} = c(\vartheta)$ is obtained by evaluating the resolvents at these two magnitudes. For a regular m-wave planform with common cross-coupling c_*,

$$\rho^2 = -\frac{a_r}{b_r + (m-1)c_{*,r}}, \qquad \lambda_{\mathrm{sym}} = -2a_r, \qquad \lambda_{\mathrm{tr}} = 2\rho^2(b_r - c_{*,r}). \tag{165}$$

The transverse eigenvalue λ_{tr} has multiplicity $m-1$. For a single active traveling direction, the local conditions reduce to

$$\rho^2 = -\frac{a_r}{b_r} > 0, \qquad -2a_r < 0, \qquad 2a_r\left(1 - \frac{c_r}{b_r}\right) < 0. \tag{166}$$

Thus $a_r > 0$, $b_r < 0$, and $c_r < b_r$ give a supercritical traveling wave that is stable against invasion by its counterpropagating partner. With $d \geq 2$, however, the isotropic critical sphere contains every direction. If $c_r(\vartheta)$ is the real cross-coupling at angular separation ϑ, stability of a single active direction against all other critical directions requires

$$\nu(\vartheta) = 2a_r\left(1 - \frac{c_r(\vartheta)}{b_r}\right) < 0 \qquad (0 < \vartheta \leq \pi). \tag{167}$$

Thus the one-dimensional coefficient $c = c(\pi)$ is a complete angular invasion test on $\mathbb{R}$, whose critical set is $\{-q_0, q_0\}$, but not on a plane or hyperspace. There one must verify (167) over the full angular continuum.

On the infinite line, the single-wave envelope can be written in a comoving and corotating frame as

$$\partial_T A = aA + D\partial_{XX}A + b|A|^2 A, \qquad D = -\frac{1}{2}\partial_q^2 \lambda_*(q_0, 0). \tag{168}$$

Put $D = D_r + \mathrm{i}D_i$ and $b = b_r + \mathrm{i}b_i$. The uniform traveling state is

$$A_*(T) = \rho \mathrm{e}^{\mathrm{i}\Omega T}, \qquad \rho^2 = -\frac{a_r}{b_r}, \qquad \Omega = a_i + b_i\rho^2. \tag{169}$$

Write $A = A_*(1 + u + \mathrm{i}v)$ and take a sideband proportional to $\mathrm{e}^{\mathrm{i}\kappa X}$. Direct linearization gives

$$\partial_T \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2\rho^2 b_r - D_r\kappa^2 & D_i\kappa^2 \\ 2\rho^2 b_i - D_i\kappa^2 & -D_r\kappa^2 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}. \tag{170}$$

Its trace and determinant are

$$\operatorname{tr}\mathbb{J}(\kappa) = 2\rho^2 b_r - 2D_r\kappa^2, \qquad \det\mathbb{J}(\kappa) = \kappa^2\left[|D|^2\kappa^2 - 2\rho^2(b_r D_r + b_i D_i)\right]. \tag{171}$$

The two-dimensional Hurwitz criterion is therefore necessary and sufficient for every nonzero sideband:

$$D_r > 0, \qquad b_r < 0, \qquad b_r D_r + b_i D_i < 0. \tag{172}$$

Equivalently, when $D_r b_r \neq 0$, the Benjamin–Feir number must satisfy

$$\mathcal{B} = 1 + \frac{D_i}{D_r}\frac{b_i}{b_r} > 0. \tag{173}$$

The continuous-wavevector calculation changes the periodic-box numbers slightly. If χ is the common coupling scale, the three-population infinite-line onset is

$$\begin{aligned} \chi_c &= 0.9999159, & q_0 &= 0.380160\,(0.1\,\mathrm{mm})^{-1}, \\ f_0 &= 14.1042\,\mathrm{Hz}, & s(0,0) &= -0.0017022\,\mathrm{ms}^{-1}, \\ \partial_q^2 \operatorname{Re}\lambda_* &= -0.0642473. \end{aligned} \tag{174}$$

Its continuous-onset coefficients and longitudinal diffusion are

$$\begin{aligned} b_3^\infty &= -0.073616 - 0.172897\mathrm{i}, & c_3^\infty(\pi) &= -1.485777 - 0.626915\mathrm{i}, \\ D_3 &= 0.032124 - 0.024065\mathrm{i}, & \mathcal{B}_3 &= -0.75944. \end{aligned} \tag{175}$$

At $a_r = 0.01$, its local amplitude is $\rho = 0.36856$ and its counterpropagating invasion rate is -0.38365. It passes the local amplitude test but fails (172); the selected traveling wave is longitudinally sideband-unstable on the infinite line at cubic order.

For the five-population model, the corresponding values are

$$\begin{aligned} \chi_c &= 1.0390375, & q_0 &= 0.491786\,(0.1\,\mathrm{mm})^{-1}, \\ f_0 &= 13.7341\,\mathrm{Hz}, & s(0,0) &= -0.0072509\,\mathrm{ms}^{-1}, \\ \partial_q^2 \operatorname{Re}\lambda_* &= -0.1153424. \end{aligned} \tag{176}$$

and

$$\begin{aligned} b_5^\infty &= -0.778597 + 0.066322\mathrm{i}, & c_5^\infty(\pi) &= -2.537480 - 0.379713\mathrm{i}, \\ D_5 &= 0.057671 + 0.002692\mathrm{i}, & \mathcal{B}_5 &= 0.99602. \end{aligned} \tag{177}$$

At $a_r = 0.01$, its local amplitude is $\rho = 0.11333$ and its counterpropagating invasion rate is -0.04518. It satisfies the complete one-dimensional cubic test: finite-wave onset, supercritical saturation, counter-wave exclusion, and longitudinal sideband stability.

A numerical scan of the angular coefficient gives

$$\max_{0<\vartheta\le\pi} c^{\infty}_{3,r}(\vartheta) = -0.14723 < b^{\infty}_{3,r}, \qquad \max_{0<\vartheta\le\pi} c^{\infty}_{5,r}(\vartheta) = -1.23240 < b^{\infty}_{5,r}. \tag{178}$$

Thus both models pass the computed angular invasion test, although only the five-population model also passes the longitudinal sideband test. This scan is numerical evidence, not an analytic bound over the angular continuum. For an isotropic plane or hyperspace, rotational degeneracy makes the real quadratic curvature vanish in directions tangent to the critical sphere:

$$\xi^T \nabla_q^2 \operatorname{Re}\lambda_*(q_0)\xi = \partial_r^2 \operatorname{Re}\lambda_*(q_0)(\xi\cdot\widehat{q}_0)^2. \tag{179}$$

Consequently full stability in $d \ge 2$ additionally requires the angular amplitude problem or the higher-order transverse phase equation. The present coefficients support, but do not yet prove, a stable isotropic plane or hyperspace pattern. An anisotropic isolated maximum can instead be decided directly by (158)– (160) when its diffusion tensor has strictly damping real part.

5.3 Extension 3: one periodic and several non-periodic dimensions

Let

$$(\theta, x) \in \mathbb{S}^1 \times \mathbb{R}^d, \qquad 0 \le \theta < 2\pi, \qquad x \in \mathbb{R}^d, \tag{180}$$

and let ℓ_θ convert angular separation to spatial units. With an orientation width σ^θ_{ij} and spatial covariance C^x_{ij}, the normalized product kernel has symbol

$$\begin{aligned} &e_{n,k}(\theta, x) = \mathrm{e}^{\mathrm{i}(n\theta + k\cdot x)}, && n \in \mathbb{Z}, \quad k \in \mathbb{R}^d, \\ &\widehat{J}(n,k)_{ij} = \mu_{ij} \exp\left[-\frac{1}{2}\left(\frac{(\sigma^\theta_{ij})^2 n^2}{\ell_\theta^2} + k^T C^x_{ij} k\right)\right]. \end{aligned} \tag{181}$$

Using one width matrix Σ means $\sigma^\theta_{ij} = \sigma_{ij}$ and $C^x_{ij} = \sigma^2_{ij} I_d$. The modal matrix and exact spectral condition are

$$M(n,k) = \mathsf{T}^{-1}\left[-I + D_1 \widehat{J}(n,k)\right], \tag{182}$$

$$\sup_{n\in\mathbb{Z},\, k\in\mathbb{R}^d} \max_s \operatorname{Re}\lambda_s(M(n,k)) < 0. \tag{183}$$

The corresponding operator decomposition is

$$\mathcal{F}_x \mathcal{L} \mathcal{F}_x^{-1} = \bigoplus_{n\in\mathbb{Z}} \int_{\mathbb{R}^d}^{\oplus} M(n,k)\, dk, \qquad \operatorname{spec}(\mathcal{L}) = \overline{\bigcup_{n\in\mathbb{Z},\, k\in\mathbb{R}^d} \operatorname{spec}(M(n,k))}. \tag{184}$$

Assume a finite critical set (n_α, k_α) with simple smooth eigenvalue bands. After removing the group velocity in the non-periodic directions, the leading expansion is

$$u_i(\theta, x, t) = u_i^* + 2\varepsilon \operatorname{Re}\sum_{\alpha=1}^{m} A_\alpha(X,T)\Phi_{i\alpha}\mathrm{e}^{\mathrm{i}(n_\alpha\theta + k_\alpha\cdot x + \omega_\alpha t)} + O(\varepsilon^2), \qquad X = \varepsilon x, \quad T = \varepsilon^2 t. \tag{185}$$

The quadratic selection equations are

$$n_r + n_s = n_\alpha, \qquad k_r + k_s = k_\alpha, \qquad \omega_r + \omega_s = \omega_\alpha, \tag{186}$$

and the cubic selection equations are

$$n_\beta + n_\gamma - n_\delta = n_\alpha, \qquad k_\beta + k_\gamma - k_\delta = k_\alpha, \qquad \omega_\beta + \omega_\gamma - \omega_\delta = \omega_\alpha. \tag{187}$$

Substitution of (n, k) for q in (120)–(121) gives all local mixed-domain coefficients. Denote that local polynomial by $\mathcal{N}_\alpha^{\mathrm{mix}}(A, \overline{A})$ and let

$$H_\alpha^x = \nabla_k^2 \lambda_\alpha(n_\alpha, k_\alpha). \tag{188}$$

The non-periodic directions add the envelope derivatives

$$\partial_T A_\alpha = \mathcal{N}_\alpha^{\mathrm{mix}}(A, \overline{A}) - \frac{1}{2} \sum_{r,s=1}^{d} (H_\alpha^x)_{rs} \partial_{X_r X_s} A_\alpha. \tag{189}$$

Constant-envelope equilibria are the roots of $\mathcal{N}^{\mathrm{mix}} = 0$. Their full stability is obtained from (158)–(159) with H_α replaced by H_α^x. This separates the discrete orientation selection rule from continuous cortical sidebands.

The mixed domain has the same onset conditions as (147)–(148) after replacing q by (n, k) and differentiating only in the continuous variable k. The critical label must satisfy $(n_0, k_0) \neq (0, 0)$, while the homogeneous sector $(0, 0)$ remains stable. If orientation is axial, $\theta \equiv \theta + \pi$, only even n occur on the 2π circle.

6 A three-population V1 example with broad excitation

6.1 What is measured and what is calibrated

Billeh et al. constructed a data-driven mouse V1 model with Gaussian distance-dependent connection probabilities. Their kernel convention was $\exp(-r^2/s^2)$, so the standard deviation in (108) is $s/\sqrt{2}$. The reported or assumed local widths used here are

Connection	s (μm)	standard deviation $s/\sqrt{2}$ (μm)
E $\to$ E	114	80.6
E $\to$ PV	92	65.1
E $\to$ SST	103	72.8
PV $\to$ E	95	67.2
SST $\to$ E	85	60.1
inhibitory $\to$ inhibitory	120	84.9

Thus E–E is broader than direct PV–E and SST–E in this local mouse-V1 dataset. This supports the concern that the broad-inhibition assumption of a two-population lateral-inhibition model should not be read literally.

The population weights below are dimensionless Wilson–Cowan gains calibrated to put the measured width ratios near a finite-wave Hopf point. They are *not* converted synaptic conductances:

$$W = \begin{pmatrix} 22.1035 & -14.3515 & -10.3472 \\ 18.1160 & -8.9407 & -12.8206 \\ 9.4942 & -2.5978 & -7.3034 \end{pmatrix}, \tag{190}$$

with rows (E, PV, SST) as targets and columns (E, PV, SST) as sources. The time constants are $(10, 12, 35)$ ms, the sigmoid equilibrium is $\overline{u} = (0.2, 0.2, 0.2)$, and the cortical patch is 4.760 mm long. Biases co-vary with the common coupling scale so this equilibrium stays fixed.

The first crossing occurs at coupling scale 1.0000165 and Fourier mode 3:

$$q_0 = 0.39598\,(0.1\,\mathrm{mm})^{-1}, \qquad \omega_0 = 0.089403\,\mathrm{ms}^{-1}, \qquad f_0 = 14.2289\,\mathrm{Hz}. \tag{191}$$

The cortical wavelength is $4.760/3 = 1.587$ mm. At the same point, the homogeneous mode has growth rate -0.001695 ms^{-1}, so the instability is genuinely spatial rather than a homogeneous Hopf instability.

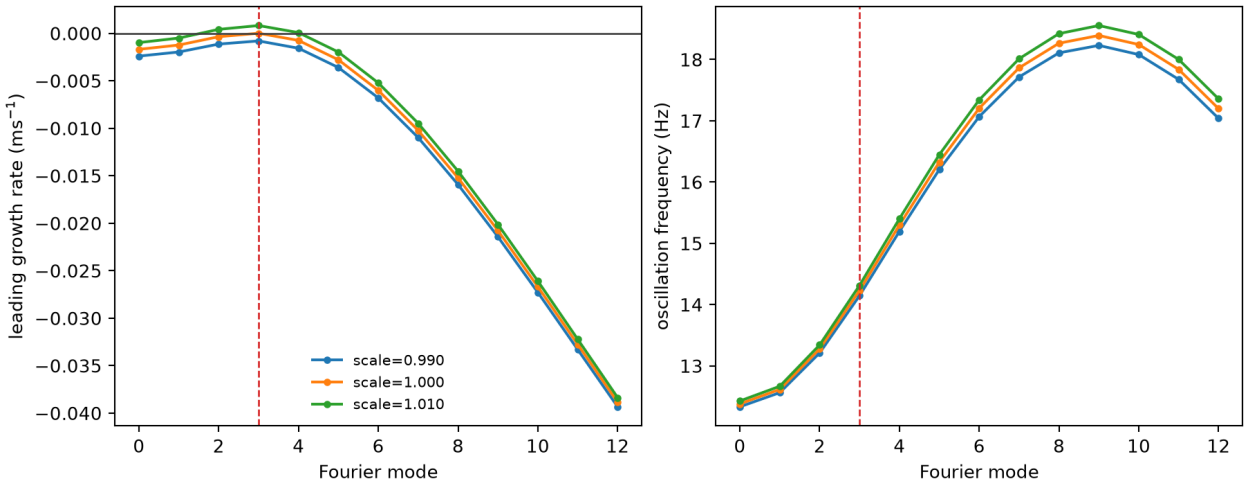


Figure 2: Leading eigenvalue across spatial modes. Mode 3 crosses first while mode 0 remains damped.

The coefficient calculation from (130)–(133) gives (3). In particular, the broad E–E connection does not prevent pattern formation once PV and SST are dynamically distinct. The mechanism uses the full three-dimensional eigenstructure and cannot be reduced to the sign of one difference of Gaussians.

6.2 What biological data justify

Three levels of evidence establish that V1 is more than a simple E/I split.

1. Tasic et al. identified 111 transcriptomic types in mouse VISp. Gouwens et al. measured 1,938 visual-cortical neurons and identified 17 electrophysiological, 38 morphological, and 46 combined morpho-electric types. Billeh et al. retained 17 coarse classes in a model of about 230,000 mouse-V1 neurons.

2. Adult mouse visual-cortex recordings show cell-type-specific connection motifs among morphologically defined interneurons. Large-scale mouse and human recordings further show that inhibitory synaptic dynamics align with the presynaptic subclass. Collapsing PV, SST, and VIP/Htr3a into one inhibitory variable discards exactly the differences that can change a Hopf instability.

3. Horizontal axon collaterals of layer 2/3 pyramidal neurons extend for millimeters in tree-shrew V1 and become feature-specific outside roughly 500 μm. Therefore broad and anisotropic

excitation is anatomically plausible, especially in species with orientation maps. A single isotropic Gaussian is already a major simplification.

The rigorous conclusion is limited but useful: a model with broad excitation, several inhibitory time scales, and cell-class-specific connections can have a stable homogeneous state and a finite-wave oscillatory instability with plausible cortical spatial and temporal scales. The present data do not identify the Wilson–Cowan gain matrix uniquely, and they do not establish that three populations are necessary. Necessity would require ruling out all two-population kernels and nonlinearities, which the current evidence cannot do.

7 Cell-type and connection audit, then a five-population test

7.1 The word "all" depends on measurement resolution

The finest transcriptomic inventory and the finest complete dynamical model are not the same object. The Allen VISp/ALM SMART-seq taxonomy contains 133 types: 56 glutamatergic, 61 GABAergic, and 16 non-neuronal types [7]. These identities do not come with a measured all-pairs physiological connection matrix. At the next coarser scale, Patch-seq and anatomical work distinguish excitatory IT, ET, NP, and CT families and inhibitory Pvalb, Sst, Vip, Lamp5, and Sncg families. Still finer names can be useful for targeting, but they cannot be inserted into (109) until their density, time scale, gain, and source–target coupling are known.

The Billeh model is presently the useful complete intermediate scale. Its 17 populations and cell counts are

Layer	excitatory	Pvalb	Sst	Htr3a
L1	–	–	–	999
L2/3	12,689	640	464	1,107
L4	10,254	963	553	270
L5	7,569	613	555	117
L6	12,882	1,052	1,059	192

Its published 17×17 table reports connection probability at a planar separation of $75\,\mu$m. The machine-readable transcription is `data/billeh_connection_probabilities_75um.csv`. A blank remains "undetermined"; it is not converted to a measured zero. Billeh et al. reported that 89 of 289 entries were undetermined and set them to zero when instantiating the circuit [6]. Probability alone is not the field weight W_{ij}: unitary PSP, convergence, population density, firing gain, delay, and synaptic dynamics must also be combined.

Several reliable motifs survive across resolutions. Pvalb cells provide fast perisomatic inhibition and strong Pvalb–Pvalb coupling. Sst cells act more slowly on distal dendrites, avoid some within-Sst connections, and inhibit several other classes. Vip/Htr3a cells include disinhibitory cells that preferentially target Sst. Excitatory connections depend on distance and functional similarity. The MICrONS volume adds more than 0.5 billion synapses across more than 200,000 cells; a proofread column resolved 1,352 neurons and more than 70,000 inhibitory synapses, including 18 connectivity motif groups and widespread interneuron target specificity [12, 13]. This evidence argues for testing several resolutions, not for claiming a completed fine-type matrix.

7.2 Application after splitting SST-GER from local SST

Drinnenberg et al. found that a median 66.1% of labeled deep-L2/3 SST cells met their general ensemble-response criterion, while 33.9% did not [14]. The following five-population test preserves

that split. Rows are targets and columns are sources in the order (E, PV, S_G, S_L, VIP):

$$W_5 = \begin{pmatrix} 22.1035 & -14.3515 & -6.8395 & -3.5077 & -1.2000 \\ 18.1160 & -8.9407 & -8.4744 & -4.3462 & -1.5000 \\ 12.5000 & -2.5978 & -4.8276 & -2.4759 & -6.0000 \\ 7.5000 & -2.5978 & -4.8276 & -2.4759 & -6.0000 \\ 8.0000 & -2.0000 & -3.3050 & -1.6950 & -1.0000 \end{pmatrix}. \tag{192}$$

The Billeh width ratios are retained except for E input to SST-GER, whose standard deviation is set to 175 μm, inside the 125–225 μm range explored by the preprint's broad-integration model. The time constants are $(10, 12, 45, 25, 20)$ ms. The SST split and width are data-constrained; the five-population gains in (192) are calibrated hypotheses, not measured conductances.

The first instability occurs at scale 1.03988, mode 4, wavelength 1.190 mm, and frequency 14.038 Hz. The homogeneous growth rate is still -0.00719 ms^{-1}. Projection gives (4), hence $\mathrm{Re}\, c_5 < \mathrm{Re}\, b_5 < 0$ and one traveling wave is locally selected. For $a = 0.01$, $\eta_t = \eta_x = 0.04$, and zero detuning, the full complex equal-amplitude state has $r = 0.14949$, $\phi = 172.622°$ modulo 180°, and Jacobian eigenvalues

$$(-0.17302, -0.08170, -0.08000, -0.07861), \tag{193}$$

so it is asymptotically stable.

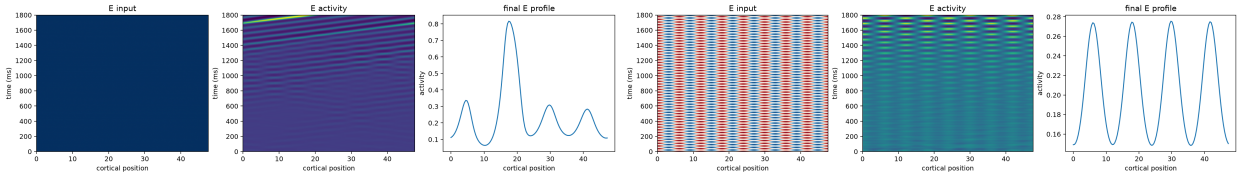


Figure 3: Five-population field. Left: above onset, one counter-propagating mode dominates. Right: below onset, combined 2:1 forcing makes the two critical amplitudes nearly equal and pins the response.

8 Hallucination patterns explained by the model

Let retinal polar coordinates be (r, θ) and approximate the retino-cortical map by

$$X = \log(r/r_0), \qquad Y = \theta. \tag{194}$$

A cortical plane wave becomes the visual-field image

$$P(r, \theta) = \cos(k_X \log(r/r_0) + k_Y \theta + \varphi)\,. \tag{195}$$

Consequently

Cortical planform	Visual-field form	Common report
$\cos(kX)$	concentric rings	target, tunnel, funnel
$\cos(nY)$	radial spokes	pinwheel, wheel, rays, cobweb-like spokes
$\cos(kX + nY)$	logarithmic spiral	spiral, rotating or rocking form
three waves at 120°	distorted lattice	honeycomb, grid, checkerboard
irregular band-limited waves	mottled or labyrinthine form	ripples, lines, zigzags, TV-like structure

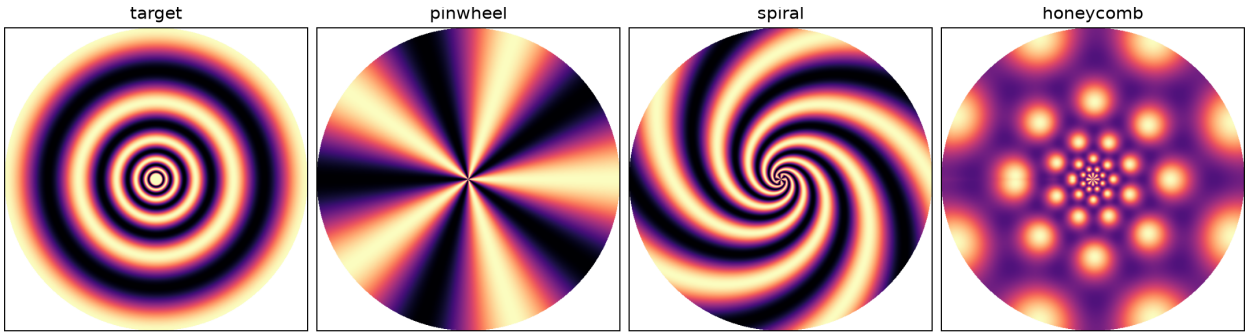


Figure 4: Static model percepts obtained by applying the inverse logarithmic retino-cortical map. These are explanatory images, not flickering stimuli.

These classes are observed outside mathematics. Allefeld et al. exposed 20 participants to a spatially uniform 1–50 Hz Ganzfeld scan. They retained 168 recurring percepts and classified lines, ripples, tunnels, honeycombs, wheels, rasters, and spirals; individual percept frequencies were often stable across repeated scans, but frequency did not determine a unique percept. The multistability of symmetry-related amplitude states is consistent with that ambiguity. A 2023 experiment with 20 screened participants compared 3, 8, 10, and 18 Hz rhythmic and arrhythmic stimulation and found the strongest geometric forms and dynamics at 10 Hz rhythmic flicker. The 2011 model produced primarily stripes at high forcing frequency and hexagons at low forcing frequency, with two frequency islands. These comparisons support the model class, but they do not prove that every reported pattern originates in V1. Color, complex imagery, retinal phosphenes, thalamic effects, and higher visual areas require additional mechanisms.

9 Numerical implementation and results

The file `neural_field.py` accepts any number of populations, a signed connectivity matrix, a Gaussian width for every source–target pair, time constants, sigmoid gains, a homogeneous rest state, dimension, boundary condition, and weak forcing. It provides:

- 1D and 2D simulations;
- periodic, reflecting, and zero exterior boundaries;
- direct 1:1, temporal 2:1, spatial 2:1, and combined weak forcing;
- automatic critical-mode and Gaussian normal-form coefficients;
- Floquet exponents of every spatial mode about a forced uniform orbit;
- exact four-real-dimensional amplitude integration and Jacobians;
- static retino-cortical inputs and numerical temporal-drive arrays.

Periodic convolutions use Fourier multipliers $\exp(-\sigma^2|q|^2/2)$. Reflecting and zero-boundary simulations use normalized Gaussian filtering in physical space. Euler time stepping is used deliberately so the code remains transparent; $\Delta t = 0.5$ ms is small relative to the shortest 10 ms population time constant.

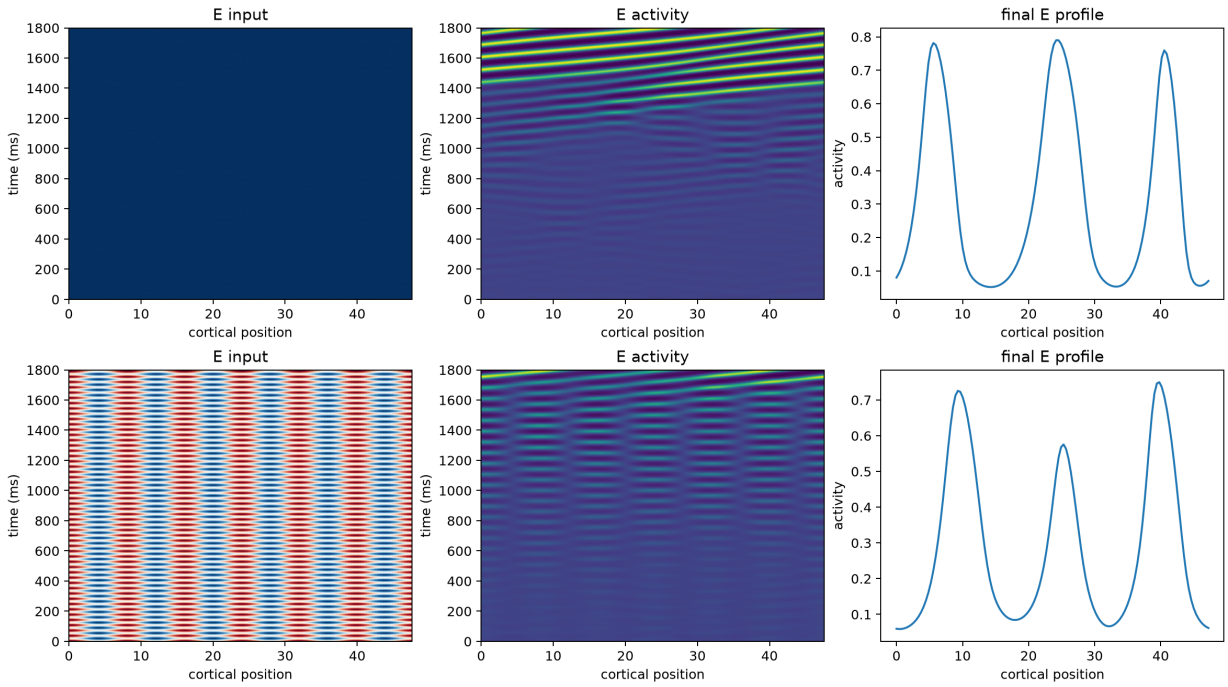


Figure 5: Top: above onset, one traveling mode dominates, as predicted by $\operatorname{Re} c < \operatorname{Re} b$. Bottom: below onset, combined temporal and spatial 2:1 forcing produces a standing, spatially pinned response.

The periodic unforced run above onset has late projected amplitudes $\langle |A| \rangle = 0.0103$ and $\langle |B| \rangle = 0.1838$, so one direction dominates. With spatial 2:1 forcing below onset, the two means are 0.0800 and 0.0681, much closer to the standing-wave diagonal. Reflecting boundaries reduce the late E standard deviation from 0.1229 in the periodic combined-forcing run to 0.0773 and visibly pin the edge response. The 2D runs form banded and labyrinthine finite-wavenumber activity rather than a homogeneous oscillation. Exact values for every run are in `output/simulations/summary.json`.

10 Model inputs and the model-predicted frequency range

The five-population intrinsic frequency is $f_0 = 14.038$ Hz. Direct and 2:1 resonant guesses are therefore 14.038 and 28.076 Hz, but a guessed center frequency is not a range. To calculate the range from the full field, let $\overline{u}_F(t)$ be the spatially uniform periodic solution under a uniform sinusoidal input of frequency F. A perturbation of spatial mode q satisfies

$$\dot{v}_q = \mathsf{T}^{-1} \left[-I + D_1(t) \widehat{W}(q) \right] v_q, \qquad D_1(t) = \operatorname{diag} F_i'(\overline{h}_{F,i}(t)). \tag{196}$$

If $P_q(F)$ is the fundamental matrix after one forcing period T_F, define

$$\gamma_q(F) = \frac{1}{T_F} \log \rho(P_q(F)). \tag{197}$$

The uniform orbit is unstable to a pattern exactly when $\max_{q \neq 0} \gamma_q(F) > 0$. This calculation includes the time-periodic sigmoid gain, switches among competing spatial modes, and requires no fitted normal-form forcing coefficient.

Figure 6 uses the five-population field at 0.99 of critical coupling. At model-input amplitude 0.025, positive finite-mode growth occurs quasi-statically below approximately 8.8 Hz, over the resolved bands 9.6–13.2 Hz and a narrow neighborhood of 14.0 Hz, and over two 2:1 tongues near 25.0–26.6 Hz and 27.2–28.6 Hz. Boundaries are grid estimates from 0.5 Hz spacing and change with coupling, input amplitude, sigmoid operating point, and the population parameters.

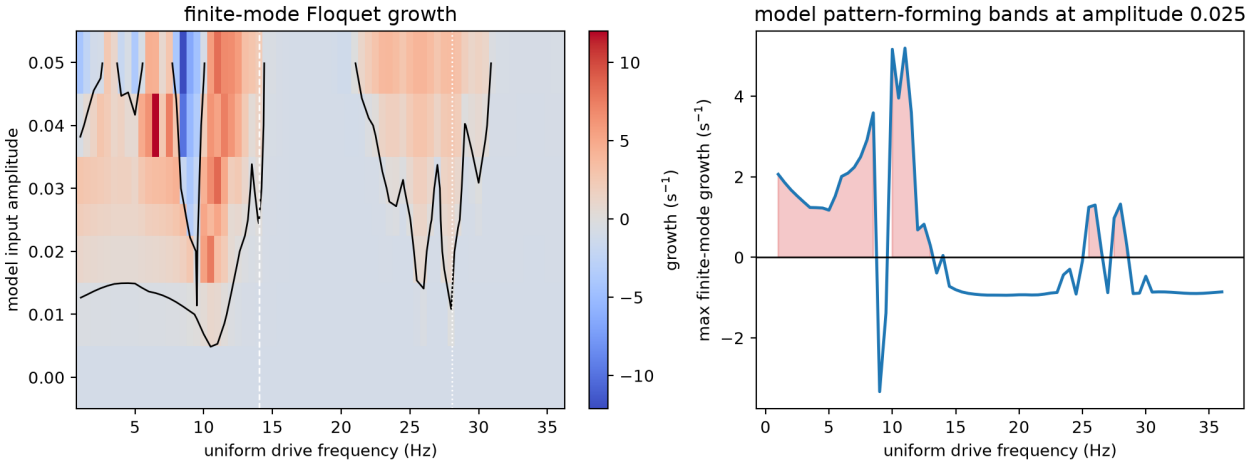


Figure 6: Left: maximum finite-mode Floquet growth versus temporal input frequency and model-input amplitude. The black contour is zero growth. Right: the amplitude-0.025 slice; red shading marks model pattern growth.

The numerical input bundle contains static target and pinwheel arrays, their PNG renderings, and numerical time series at f_0 and $2f_0$ in `output/simulations/model_inputs/`. It contains no animation or autoplay display. The computed Floquet intervals are a *model pattern-forming range*; they are not a measured human hallucination threshold and they are not a safety range. The Epilepsy Foundation identifies roughly 5–30 Hz as particularly relevant for photosensitive seizures, overlapping the computed tongues, while W3C guidance avoids more than three flashes per second. Therefore no high-contrast human-viewable flicker intended to provoke hallucinations is generated here. Static high-contrast stripes and radial patterns can also be provocative for pattern-sensitive viewers, so the static arrays are model inputs rather than validated exposure stimuli. Any human stimulation requires an approved clinical or psychophysics protocol with screening, luminance and contrast limits, field-size limits, emergency stopping, and informed consent.

11 Limitations and next experimental tests

1. The Gaussian width ratios are data-informed; the population weights in (190) are calibrated. Direct conversion from connection probability and unitary synaptic strength to a Wilson–Cowan gain requires cell density, convergence, release dynamics, operating point, and gain.

2. Mouse V1 has a salt-and-pepper orientation organization, while the classical human hallucination theory uses orientation columns. Human, primate, carnivore, and tree-shrew horizontal connectivity should be used when spatially precise human predictions are intended.

3. The cubic Turing–Hopf normal form explains the local high-frequency stripe selection. Low-frequency hexagons require the Floquet normal form about the periodically forced homogeneous orbit and its quadratic term.

4. The logarithmic map predicts geometry, not color. Color opponency needs additional cone-opponent or cortical color populations.

5. The strongest test is quantitative: measure each participant's damped visual frequency, then predict both the two forcing islands and the spatial wavelength before collecting hallucination reports.

A One-page summary: large-scale cellular-resolution read/write

Paper and question. Drinnenberg et al., "Large-scale cellular-resolution read/write of activity enables discovery of cell types defined by complex circuit properties," is a 2025 bioRxiv preprint and has not been peer reviewed [14]. The paper asks whether a cell type can be defined by how it participates in a large causal circuit experiment, rather than only by genes, morphology, or intrinsic physiology. The central logic is to stimulate many precisely chosen excitatory ensembles, read thousands of non-targeted neurons, identify followers common to otherwise non-overlapping ensembles, then use cell labels and a probabilistic model to explain that response class.

Read/write experiment. New transgenic and follower-focused preparations combined soma-enriched ChRmine with calcium indicators. The platform controlled approximately 10^3 cells and monitored up to 10^4 cells within or across mouse visual areas. In the main V1 experiment, 104 random, orientation-selective, or scene-selective L2/3 excitatory ensembles were tested in 10 awake mice. Each ensemble had a median 50 targets and was driven for 1.5 s at 32 Hz while a median 8,318 non-targeted regions of interest were monitored. Downstream activity decoded ensemble identity above chance; iso-tuned cells were recruited while orthogonally tuned cells were suppressed, consistent with like-to-like wiring.

Probabilistic model. For a candidate follower, the connection probability from a pyramidal cell at distance r was

$$p(r) = A \exp[-r^2/(2\sigma^2)]. \tag{198}$$

If p_c is this probability averaged over uniformly distributed targets, the probability of following an ensemble of T targets with activation threshold k is

$$P_f = 1 - \sum_{j=0}^{k-1} \binom{T}{j} p_c^j (1-p_c)^{T-j}, \qquad \mathbb{E}[m] = M P_f \tag{199}$$

for M independently sampled ensembles. With $M = 10$, $T = 50$, peak connectivity $A = 0.2$, $\sigma = 100\,\mu$m, and $k = 5$, the expected number followed was only 0.004 of 10. Larger σ helped, but a low k was also necessary. The model therefore predicted a broadly integrating, low-threshold inhibitory subtype rather than an ordinary local follower.

Results. Single ensembles recruited 199 ± 9 positive followers (2.5% of the non-targeted population) and suppressed 267 ± 14 cells (3.4%). Of the positive followers, $18 \pm 3.8\%$ followed at least four ensembles, over 20-fold the chance expectation $0.87 \pm 0.3\%$; $6.3 \pm 1.8\%$ followed at least eight. These general ensemble-response (GER) neurons remained active more than $150\,\mu$m from the nearest target, were concentrated in deep L2/3, were broadly visually responsive, and were mutually correlated. Cell-type labeling showed PV neurons were suppressed, whereas SST neurons were increased. A median 66.1% of labeled SST cells met the GER criterion and 33.9% did not; 53.5% of all functionally identified GER cells carried the incomplete SST label, so the true SST fraction may be higher.

Connection to this model. The result supports splitting SST-GER from local SST: the former receives a broader E kernel, has a lower effective threshold and longer integration time, and supplies dendritic feedback inhibition. It does not provide a measured five-by-five gain matrix. Equation (192) is therefore a transparent hypothesis test. Its finite-wave Hopf point and coefficients show that the functional split is sufficient to preserve pattern formation with broad excitation; they do not establish SST-GER dysfunction as a human hallucination mechanism.

References

[1] M. Rule, M. Stoffregen, and B. Ermentrout, "A Model for the Origin and Properties of Flicker-Induced Geometric Phosphenes," *PLoS Computational Biology* 7, e1002158 (2011). doi:10.1371/journal.pcbi.1002158.

[2] P. Bressloff, J. Cowan, M. Golubitsky, P. Thomas, and M. Wiener, "What Geometric Visual Hallucinations Tell Us about the Visual Cortex," *Neural Computation* 14, 473–491 (2002). doi:10.1162/089976602317250861.

[3] C. Allefeld, P. Putz, K. Kastner, and J. Wackermann, "Flicker-light induced visual phenomena: Frequency dependence and specificity of whole percepts and percept features," *Consciousness and Cognition* 20, 1344–1362 (2011). doi:10.1016/j.concog.2010.10.026.

[4] M. Bartossek et al., "Effect of frequency and rhythmicity on flicker light-induced hallucinatory phenomena," *PLoS ONE* 18, e0284271 (2023). doi:10.1371/journal.pone.0284271.

[5] V. Billock and B. Tsou, "Neural interactions between flicker-induced self-organized visual hallucinations and physical stimuli," *PNAS* 104, 8490–8495 (2007). doi:10.1073/pnas.0610813104.

[6] Y. Billeh et al., "Systematic Integration of Structural and Functional Data into Multi-scale Models of Mouse Primary Visual Cortex," *Neuron* 106, 388–403 (2020). doi:10.1016/j.neuron.2020.01.040.

[7] B. Tasic et al., "Shared and distinct transcriptomic cell types across neocortical areas," *Nature* 563, 72–78 (2018). doi:10.1038/s41586-018-0654-5.

[8] N. Gouwens et al., "Classification of electrophysiological and morphological neuron types in the mouse visual cortex," *Nature Neuroscience* 22, 1182–1195 (2019). doi:10.1038/s41593-019-0417-0.

[9] X. Jiang et al., "Principles of connectivity among morphologically defined cell types in adult neocortex," *Science* 350, aac9462 (2015). doi:10.1126/science.aac9462.

[10] L. Campagnola et al., "Local Connectivity and Synaptic Dynamics in Mouse and Human Neocortex," *Science* 375, eabj5861 (2022). doi:10.1126/science.abj5861.

[11] W. Bosking, Y. Zhang, B. Schofield, and D. Fitzpatrick, "Orientation Selectivity and the Arrangement of Horizontal Connections in Tree Shrew Striate Cortex," *Journal of Neuroscience* 17, 2112–2127 (1997). doi:10.1523/JNEUROSCI.17-06-02112.1997.

[12] MICrONS Consortium, "Functional connectomics spanning multiple areas of mouse visual cortex," *Nature* 640, 435–447 (2025). doi:10.1038/s41586-025-08790-w.

[13] C. Schneider-Mizell et al., "Inhibitory specificity from a connectomic census of mouse visual cortex," *Nature* 640, 448–458 (2025). doi:10.1038/s41586-024-07780-8.

[14] A. Drinnenberg et al., "Large-scale cellular-resolution read/write of activity enables discovery of cell types defined by complex circuit properties," bioRxiv (2025). doi:10.1101/2025.10.21.683734.

[15] Epilepsy Foundation, "Photosensitivity and Seizures." epilepsy.com.

[16] W3C Web Accessibility Initiative, "Understanding Success Criterion 2.3.2: Three Flashes." w3.org.